Module 5: Pavement Design (8 Lectures)

- Basic Principles, Methods for different Types of Pavements, Design of flexible pavement using IRC: 37- 2012, Design of rigid pavement using IRC: 58-2011

Other modes of Transport

- Introduction to Railways, Airways, Waterways, Pipeline Transportation, Classification, Requirements, Comparative Studies

Pavement Design Basic Principle

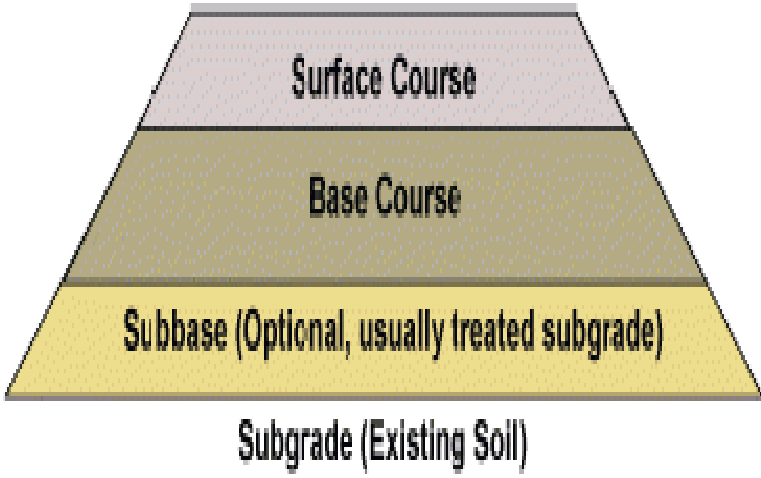
S.No	Criteria	Flexible Pavement (IRC:37-2018)	Rigid Pavement (IRC:58-2015)
1	**Design Basis**	Empirical + Mechanistic-Empirical Approach	Mechanistic Design Approach
2	**Load Consideration**	Traffic in terms of cumulative standard axles (msa)	Traffic in terms of cumulative standard axles (msa)
3	**Traffic Input**	Initial Traffic (CVPD), Traffic Growth Rate, Design Life	Same as Flexible Pavement
4	**Material Properties**	Elastic Modulus, Poisson's Ratio, Fatigue Properties	Flexural Strength (Modulus of Rupture), Elastic Modulus
5	**Subgrade Strength**	CBR value of Subgrade Soil	Modulus of Subgrade Reaction (k-value)
6	**Environmental Factors**	Temperature Variation, Drainage Conditions	Temperature Gradient (for warping stresses)
7	**Failure Criteria**	Fatigue Cracking, Rutting	Fatigue Cracking, Corner Breaks, Joint Failures
8	**Pavement Composition**	Surface Course, Base Course, Sub-base, Subgrade	Concrete Slab, Sub-base, Subgrade
9	**Design Life**	15 years (typical)	30 years (typical)
10	**Drainage Requirements**	Proper layer drainage and surface drainage	Efficient sub-surface drainage and joint sealing
11	**Stress/Strain Checks**	Vertical strain on Subgrade, Horizontal tensile strain on bituminous layer	Flexural stress, Load transfer efficiency at joints
12	**Layer Thickness Determination**	Based on fatigue and rutting criteria charts (IRC:37)	Based on stress analysis, thickness design charts (IRC:58)

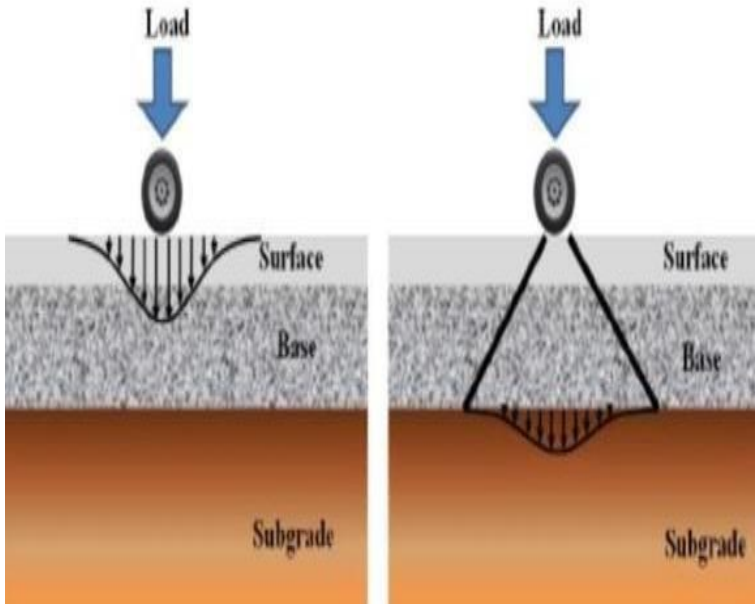
Notes:

- **CVPD:** Commercial Vehicles Per Day.
- **msa:** Million Standard Axles (cumulative over design life).
- IRC allows adjustments based on specific **climatic zones**, **material properties**, and **construction quality control**.

1. Flexible Pavements

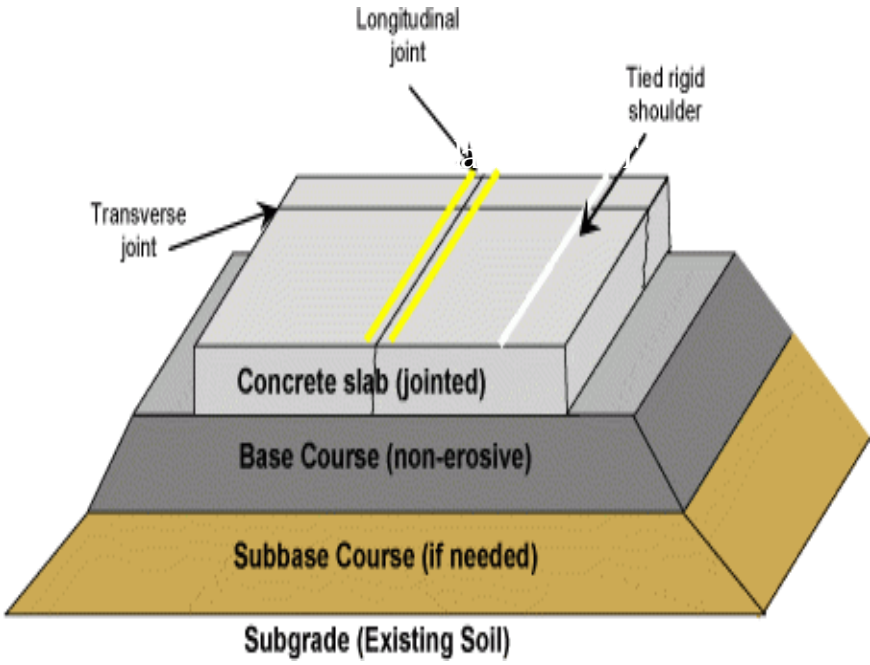
- Flexible pavement consists of a mixture of asphaltic or bituminous material and aggregates placed on a bed of compacted granular material of appropriate quality in layers over the subgrade.
- Water bound macadam roads and stabilized soil roads with or without asphaltic toppings are examples of flexible pavements.
- The **design of flexible pavement** is based on the principle that for a load of any magnitude, the intensity of a load diminishes as the **load is transmitted downwards from the surface by virtue of spreading over an increasingly larger area,** by carrying it deep enough into the ground through successive layers of granular material.



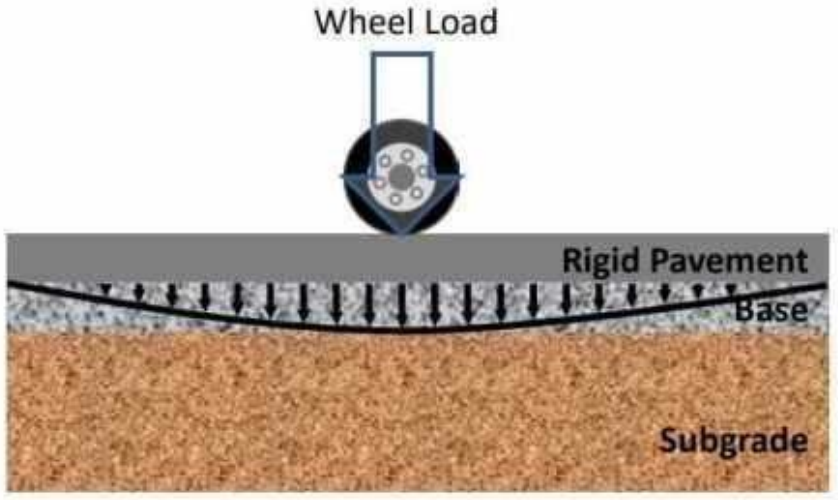


Rigid Pavements

- A rigid pavement is constructed from cement concrete or reinforced concrete slabs.
- Grouted concrete roads are in the category of semi-rigid pavements.
- The design of rigid pavement is based on providing a **structural cement concrete slab of sufficient strength** to resists the loads from traffic.



- The rigid pavement has **rigidity and high modulus of elasticity to distribute the load** over a relatively wide area of soil.



Point	Flexible Pavements	Rigid Pavements
Initial cost	Initial cost is low.	Initial cost is high.
Life span	Short life span.	Long life span.
Thickness	Thickness is more.	Thickness is less.
Joints	Joints are not required.	Joints are essentially required.
Subgrade	A reasonably good subgrade is required.	A good subgrade is not necessary.
Durability	Less durable	More durable.
Temperature effect	Stresses are not induced	Heavy stresses are induced
Repair work	Easy repair work.	Difficult to repair.
Skill and supervision	Moderate skill and less supervision are required.	More skill and supervision is required.
Feasibility of stage development	Stage development is feasible.	Stage development is not feasible.

Point	Flexible Pavements	Rigid Pavements
Openings to traffic after construction	Can be opened to traffic shortly after construction.	Require curing after construction and thus cause delay in opening to traffic.
Behavior with subgrade settlement.	They adjust according to any deformation of subgrade without rupture.	They do not adjust according to any deformation of subgrade without rupture.
Tractive resistance	More tractive resistance.	Less tractive resistance.
Traffic suitability	Suitable for all types of traffic.	Noisy under iron wheeled traffic.
Night visibility	Blacktop pavements provide poor visibility at night.	They offer good visibility at night.
Effect Of loading	They adjust themselves to normal loading by undergoing elastic deformation.	They tend to act as a beam or cantilever and resist deformation.
Maintenance cost	Maintenance cost is high.	Maintenance cost is low.
Glaring effect	They do not usually cause glare due to reflected sunlight.	They often cause glare due to reflected sunlight.

Selection of Flexible & Rigid Road Pavement :

Flexible:

1) When funds are not sufficiently available.
2) Large number of roads are to be constructed.
3) Good subgrade is available.
4) Local material are to be advantageously used.
5) Traffic load is not very heavy.
6) Pavement is to be constructed in stages.

Rigid:

1) When funds are sufficiently available.
2) Long life of the road pavement is desired.
3) Subgrade is poor & varying in nature.
4) Route is very important.
5) Traffic load is very heavy.

An **ideal pavement should meet** the following requirements:

- **Sufficient thickness to distribute the wheel load stresses** to a safe value on the sub-grade soil,
- **Structurally strong** to withstand all types of stresses imposed upon it,
- **Adequate coefficient of friction** to prevent skidding of vehicles,
- **Smooth surface** to provide comfort to road users even at high speed,
- Produce **least noise** from moving vehicles,
- **Dust proof surface** so that traffic safety is not impaired by reducing visibility,
- **Impervious surface**, so that sub-grade soil is well protected, and
- **Long design life** with low maintenance cost.

Design of flexible pavement using IRC: 37- 2012

- Indian roads congress has specified the design procedures for flexible pavements based on **CBR values.**

- The Pavement designs given in the previous edition IRC:37-1984 were applicable to design track up to only **30 million standard axles (msa).**

- The earlier code is empirical in nature which has limitations regarding applicability and extrapolation.

- This guidelines follows analytical designs and developed new set of designs up to 150 msa in IRC:37-2001.

SCOPE-

- These guidelines will apply to design of flexible pavements for Expressway, National Highways, State Highways, Major District Roads, and other categories of roads.

- Flexible pavements are considered to include the pavements which have bituminous surfacing and granular base and sub-base courses conforming to IRC/ MOST standards.

- These guidelines apply to new pavements.

Methods for Flexible Pavement Design (IRC:37-2012)

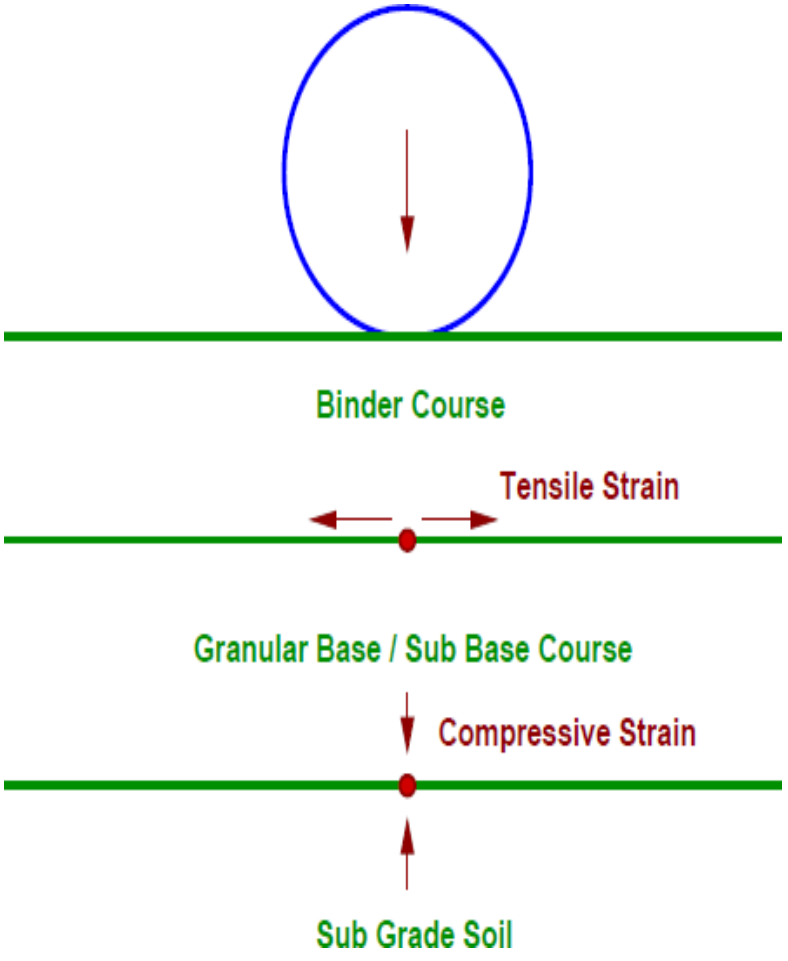
S.No	Step/Method	Description
1	Design Approach	Semi-Mechanistic Empirical Method (combination of calculated strains and observed pavement performance)
2	Performance Criteria	- Fatigue cracking: limit horizontal tensile strain at bottom of bituminous layer - Rutting: limit vertical compressive strain at top of subgrade
3	Critical Strains	- Fatigue: Horizontal tensile strain (ε_t) - Rutting: Vertical compressive strain (ε_v)
4	Layered Elastic Analysis	Use elastic layered theory (KENPAVE, IITPAVE, etc.) to calculate stresses and strains
5	Traffic Estimation	Based on cumulative standard axles (msa) over design life considering growth rate, VDF, traffic volume
6	Material Characterization	Assign Elastic Modulus (E) values for: - Bituminous Layers - Granular Base - Granular Sub-Base - Subgrade
7	**Design Charts Provided**	**IRC:37-2012 provides catalogues of layer thicknesses for different subgrade CBR and traffic (msa)**
8	Special Conditions	Adjust for temperature, weak subgrade, heavily loaded vehicles by modifying material properties or thicknesses
9	Pavement Composition	Bituminous layers + Granular base + Granular sub-base over prepared subgrade (treated if CBR < 5%)
10	Failure Criteria	Pavement is safe if both fatigue life and rutting life are greater than design traffic

DESIGN CRITERIA

- The flexible pavements has been modeled as a **three layer** structure and stresses and strains at critical locations have been computed using the **linear elastic** model.
- To give proper consideration to the aspects of performance, the following three types of pavement distress resulting from **repeated (cyclic) application** of traffic loads are considered:

1. **Vertical compressive strain** at the top of the sub-grade which can cause sub-grade deformation resulting in permanent deformation at the pavement surface.
2. **Horizontal tensile strain** or stress at the bottom of the bituminous layer which can cause fracture of the bituminous layer.
3. **Pavement deformation** within the bituminous layer.

- For calculating tensile strains at the bottom of the bituminous layer, the **stiffness of dense bituminous macadam (DBM) layer with 60/70** bitumen has been used in the analysis.
- While the permanent deformation within the bituminous layer can be controlled by meeting the mix **design requirements, thickness of granular and bituminous layers** are selected using the analytical design approach so that strains at the critical points are within the allowable limits.



Critical Locations in Pavement

Design procedure

- Based on the performance of existing designs and using **analytical approach, simple design charts and a catalogue of pavement** designs are added in the code.
- The pavement designs are given for subgrade **CBR values ranging from 2% to 10% and design traffic ranging from 1 msa to 150 msa** for an average annual pavement temperature of 35d C.
- The later thicknesses obtained from the analysis have been slightly modified to adapt the designs to stage construction.
- Using the following simple input parameters, appropriate designs could be chosen for the given traffic and soil strength:

Design traffic in terms of cumulative number of standard axles and CBR value of subgrade.

Factors affecting design and performance of flexible pavement:

1. **Wheel loads of heavy vehicles or traffic loads**
2. Subgrade soil
3. Climatic Factors
4. Pavement component material in different layers
5. Drainage and Environmental Factors
6. **Design Traffic**

1. Magnitude of wheel loads

The **wheel load** or **axle load** is the load transmitted to the pavement surface by the wheels of a vehicle. In case of **multi-axle vehicles**, the load is distributed between the axles. For pavement design, we focus on the **single axle load** as it is often the highest load.

In India, the standard **design axle load** as per **IRC** (Indian Roads Congress) is:

- **Single Axle Load (SINGLE)**
  - The **standard axle load** is typically taken as **8.16 tons** or **8160 kg**.
- **Standardized Axle Load (SAE)**:
  - This is a **Standard Equivalent Axle** to convert **different axle loads** into equivalent **8.16-ton single axle load** for **pavement design**.

1. For example, if a vehicle has a single axle load of 8.16 tons (standard axle), and if the vehicle has two wheels on each side of the axle, the wheel load per wheel can be calculated as:

   Wheel Load = Axle Load / Number of Wheels

2. If there are 2 wheels per axle, then:

   Wheel Load = 8.16 tons / 2 = 4.08 tons per wheel.

Contact Pressure:

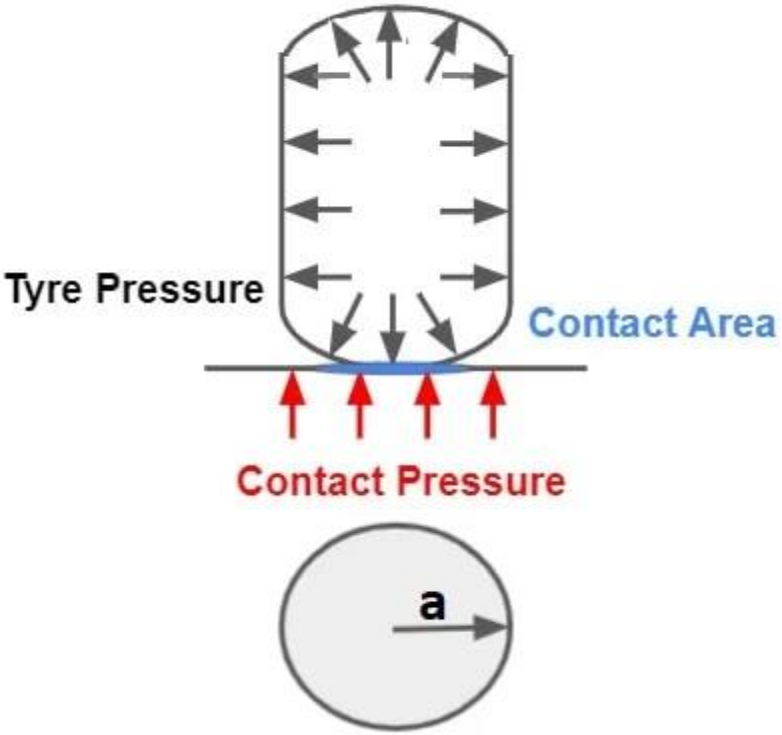
Contact pressure refers to the pressure exerted by the wheel onto the pavement surface at the area where the tire touches the ground.

The formula for calculating contact pressure is:

Contact Pressure = Wheel Load / Contact Area

The contact area depends on factors like tire size, inflation pressure, and the type of vehicle.

For a typical passenger car, the contact area is approximately 0.1 to 0.2 m².



Contact Pressure for Various Vehicle Types

Vehicle Type	Wheel Load (per wheel)	Contact Area	Contact Pressure (approx.)
Passenger Car	4–5 tons (~4000–5000 N)	0.1 – 0.2 m²	200 – 500 kPa
Truck (Single Axle)	8.16 tons (~8000 N)	0.4 – 0.6 m²	1300 – 2000 kPa
Truck (Dual Axle)	16.32 tons (~16,320 N)	0.8 – 1.0 m²	1600 – 2000 kPa
Bus (Single Axle)	6–10 tons (~6000–10,000 N)	0.4 – 0.6 m²	1000 – 2500 kPa

Equivalent Single Wheel Load (ESWL)

Definition:

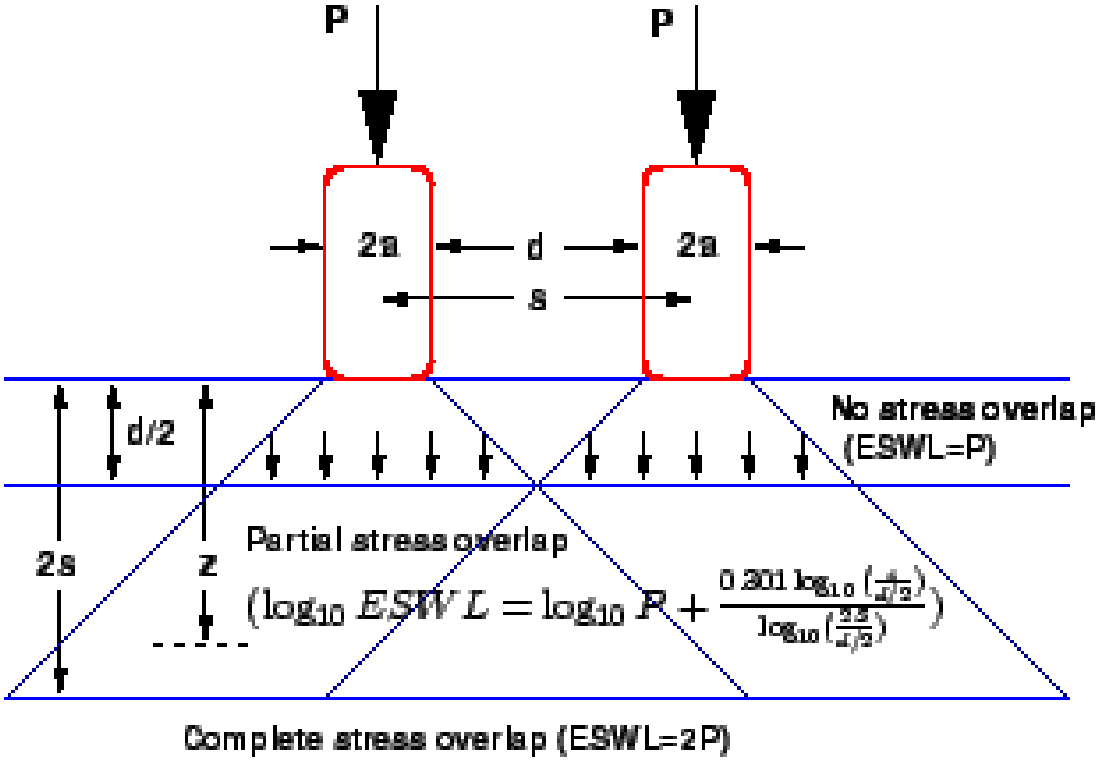
The **Equivalent Single Wheel Load (ESWL)** is the **single wheel load** that would produce the **same effect** (stress, strain, or deflection) on the pavement as that produced by a group of wheels (like dual wheels or multiple axles).

Why is ESWL Needed?

•Real vehicles have **multiple wheels** and **multiple axles**.

•It's complicated to design pavement for every wheel separately.

•So, we use **one imaginary "equivalent" wheel load** (ESWL) to **simplify the pavement design**.



Sl. No.	Axle Load (ton)	Wheel Load (ton)	Clear Distance between Tyres (cm)	ESWL at Depth 0 cm (ton)	ESWL at Depth 20 cm (ton)	ESWL at Depth 40 cm (ton)
1	8.16	4.08	31.5	4.08	3.15	2.48
2	10.2	5.1	35	5.1	3.9	3.05
3	13.0	6.5	38	6.5	5.0	3.9
4	15.2	7.6	40	7.6	5.8	4.5
5	18.0	9.0	45	9.0	6.7	5.2

How the Table is Used:

- **At surface (0 cm depth):** Each wheel acts independently → ESWL = Wheel load.
- **At 20 cm depth:** Overlap starts → ESWL reduced.
- **At 40 cm depth:** More overlap → ESWL further reduced.

Thus, **as depth increases, ESWL decreases** because the stress bulbs from dual wheels merge.

Assumptions as per IRC

Parameter	Value
Standard Axle Load	8.16 tonnes (8160 kg)
Standard Wheel Load	4.08 tonnes (per wheel)
Contact Pressure	0.56 MPa (assumed)
Tyre Pressure	~0.8 MPa (varies by vehicle)
Clear Distance between Tyres	30–45 cm (commonly)

Equivalent Standard Axle Load (ESAL)

Definition:

The **Equivalent Standard Axle Load (ESAL)** is a method used to express **different types of axle loads** (single, tandem, tridem) in terms of a **standard axle load**.

In short:

ESAL = How much damage a given axle causes compared to a standard axle.

Why is ESAL Needed?

- Real traffic has **many types** of vehicles — buses, trucks, multi-axle trailers.
- Each vehicle damages the pavement differently.
- So we **convert all vehicle loads** into **equivalent repetitions of a standard axle load**.

Thus, **design traffic** is expressed as **Million Standard Axles (MSA)**.

Formula for ESAL

The **Load Equivalency Factor (LEF)** is calculated using:

$$LEF = \left(\frac{W}{W_{std}}\right)^4$$

Where:

- W = Actual axle load (tonnes)
- W_{std} = Standard axle load = 8.16 tonnes
- 4 = Power (for flexible pavement; varies for rigid)

Thus:

$$\text{ESAL} = \text{Axle Repetitions} \times LEF$$

Design Traffic

- The method considers **Design Traffic** in terms of the cumulative number of standard axles to be carried by the pavement during the design life.
- This requires the following information:

1. Initial traffic in terms of CVPD (commercial vehicles per day)
2. Traffic growth rate during the design life
3. Design life in number of years
4. Vehicle damage factor (VDF)
5. Distribution of commercial traffic over the carriage way.

Initial traffic

- Initial traffic is determined in terms of commercial vehicles per day (CVPD). For the structural design of the pavement only commercial vehicles are considered assuming laden weight of three tonnes or more and their axle.
- loading will be considered. Estimate of the initial daily average traffic flow for any road should normally be based on 7-day 24-hour classified traffic counts (ADT).
- In case of new roads, traffic estimates can be made on the basis of potential land use and traffic on existing routes in the area.

Traffic growth rate

- Traffic growth rates can be estimated (i) by studying the past trends of traffic growth, and (ii) by establishing econometric models.
- If adequate data is not available, it is recommended that an average annual growth rate of 7.5 percent may be adopted.

Design life

- For the purpose of the pavement design, the design life is defined in terms of the cumulative number of standard axles that can be carried before strengthening of the pavement is necessary.
- It is recommended that pavements for arterial roads like NH, SH should be designed for a life of 15 years, EH and urban roads for 20 years and other categories of roads for 10 to 15 years.

Vehicle Damage Factor-

- *"Vehicle Damage Factor is the number of standard axles per commercial vehicle.“*
- In simple terms, it refers to how much damage a single vehicle causes to a road compared to a standard reference vehicle usually a standard axle load of 80 kN

Lane type	Vehicle damage factor
Single lane roads	1.00
Two-lane single carriageway roads	0.75
Four-lane single carriageway roads	0.40
Dual two carriageway roads	0.75
Dual three carriageway roads	0.60
Dual four carriageway roads	0.45

VDF Values depend on:

1. Vehicle type (2-axle, 3-axle, multi-axle trucks)
2. Load carried (heavily overloaded vehicles have **higher VDF**)
3. Road type (NH, SH, MDR)
4. Region (Plain, Hilly)

VDF = Vehicle Damage Factor-
(Total number of equivalent standard axles) / (Total number of vehicles)

Typical VDF values from IRC:37

Type of Road	Traffic Category	Terrain	VDF
National Highway	Heavy Traffic	Plain	3.5 – 5.0
State Highway	Moderate Traffic	Plain	2.5 – 3.5
Major District Roads	Low Traffic	Plain	2.0 – 3.0
All roads	Any Traffic	Hilly	1.5 – 2.5

Suppose you have a traffic survey with this data:

Vehicle Type	Number of Vehicles	Equivalent Standard Axle Load (ESAL) per vehicle
Car	10,000	0.0004
Light Commercial Van	2,000	0.01
Medium Truck	1,000	0.5
Heavy Truck	500	3.5

Step 1: Calculate Total ESAL
Total ESAL = (10,000 × 0.0004) + (2,000 × 0.01) + (1,000 × 0.5) + (500 × 3.5)
Total ESAL = (4) + (20) + (500) + (1750) = **2274**

Step 2: Calculate Total Vehicles
Total Vehicles = 10,000 + 2,000 + 1,000 + 500 = **13,500 vehicles**

Step 3: Calculate VDF
VDF = Total ESAL ÷ Total Vehicles
VDF = 2274 ÷ 13,500 ≈ **0.168**

Interpretation:
Each vehicle on average is causing **0.168 times the damage** of a standard axle load.

Vehicle distribution-

- A realistic assessment of distribution of commercial traffic by direction and by lane is necessary as it directly affects the total equivalent standard axle load application used in the design. The following distribution may be assumed-
  - ✓ **Single lane roads:** Traffic tends to be more channelized on single roads than two lane roads and to allow for this concentration of wheel load repetitions, the design should be **based on total number of commercial vehicles in both directions.**
  - ✓ **Two-lane single carriageway roads:** The design should be based on **75 % of the commercial vehicles in both directions.**
  - ✓ **Four-lane single carriageway roads:** The design should be based **on 40 % of the total number of commercial vehicles in both directions.**
  - ✓ **Dual carriageway roads:** For the design of dual two-lane carriageway roads should be **based on 75 % of the number of commercial vehicles in each direction**. For dual three- lane carriageway and dual four-lane carriageway the **distribution factor will be 60 % and 45 % respectively**.

Pavement thickness design charts-

- The design curves relate pavement thickness to the cumulative number of standard axles to be carried over the design life for different sub-grade **CBR values ranging from 2 % to 10 %.**
- The **design charts will give the total thickness of the pavement** for the above inputs.
- The **total thickness consists of granular sub-base, granular base and bituminous surfacing.**
- The individual layers are designed based on the recommendations given and the subsequent tables.
- In **IRC:37:2001**, pavement design is done using two different charts based on traffic load:

Traffic Range	Design Chart Used
1 msa to 10 msa	**Chart 1**
10 msa to 150 msa	**Chart 2**

Flexible Pavement Thickness (IRC:37-2018)

Design Traffic (msa)	Total Pavement Thickness (mm) — CBR 3%	Total Pavement Thickness (mm) — CBR 5%	Total Pavement Thickness (mm) — CBR 8%	Total Pavement Thickness (mm) — CBR 10%
1	660	510	430	390
2	700	540	460	420
5	760	600	500	460
10	820	660	560	520
20	890	720	610	560
30	950	770	660	610
50	1020	850	720	670
100	1140	960	810	760
150	1210	1030	880	820

Layer-wise Pavement Composition (as per IRC:37-2018)
(for Subgrade CBR = 5%)

Traffic (msa)	GSB Thickness (mm)	WMM/WBM Thickness (mm)	DBM Thickness (mm)	BC Thickness (mm)	**Total Thickness (mm)**
1	200	250	40	20	510
2	200	270	50	20	540
5	230	280	70	20	600
10	230	300	90	40	660
20	230	330	100	60	720
30	250	340	120	60	770
50	250	370	130	100	850
100	250	400	150	160	960

Explanations:

- **GSB** = Granular Sub-Base (minimum 200–250 mm)
- **WMM/WBM** = Granular Base (increases with msa)
- **DBM** = Dense Bituminous Macadam (bituminous base layer)
- **BC** = Bituminous Concrete (surface course)

(*Thickness increases with traffic for durability*)

Pavement composition-

Sub-base-

- sub-base materials comprise natural **sand, gravel, laterite, brick metal, crushed stone or combinations** there of meeting the prescribed grading and physical requirements.
- The sub-base material should have a **minimum CBR of 20 % and 30 % for traffic up to 2 msa and traffic exceeding 2 msa respectively.**
- Sub-base usually consist of **granular or WBM and the thickness should not be less than 150 mm for design traffic less than 10 msa and 200 mm for design traffic of 1:0 msa and above.**

Base-

- The recommended designs are for unbounded granular bases which comprise conventional water bound macadam (WBM) or wet mix macadam (WMM) or equivalent confirming to MOST specifications.
- The materials should be of good quality with minimum thickness of 225 mm for traffic up to 2 msa an 150 mm for traffic exceeding 2 msa.

Bituminous surfacing-

- The surfacing consists of a wearing course or a binder course plus wearing course.
- The most commonly used wearing courses are surface dressing, open graded premix carpet, mix seal surfacing, semi-dense bituminous concrete and bituminous concrete.
- For binder course, MOST specifies, it is desirable to use bituminous macadam (BM) for traffic upto 5 msa and dense bituminous macadam (DBM) for traffic more than 5 msa.

Flexible Pavement Design: Steps as per IRC:37-2012

1. Calculate Traffic Growth Factor (TGF)

Formula: $TGF = \frac{(1+r)^n - 1}{r}$

Where:

- r = Annual growth rate (decimal)
- n = Design life (years)

Example:

- r = 0.07 (7%)
- n = 20 years

$$TGF = \frac{(1+0.07)^{20} - 1}{0.07} \approx 38.696$$

2. Calculate the Total Number of Commercial Vehicles

Formula: $TotalVehicles = CV \times 365 \times TGF$

Example:

- CV = 5000 vehicles/day
- TGF = 38.696

$$TotalVehicles = 5000 \times 365 \times 38.696 \approx 70.65 \times 10^6$$

3. VDF = Vehicle Damage Factor- (Total number of equivalent standard axles) / (Total number of vehicles)

4. Adjust for Lane Distribution factor

Road Type / Facility	Number of Lanes	(LDF)	Notes
Single carriageway	2 lanes (undivided)	0.75	75% of traffic assumed on one lane
Dual carriageway	4 lanes (divided)	0.40	40% on the most heavily loaded lane
Dual carriageway	6 lanes (divided)	0.25	25% on the most heavily loaded lane
Dual carriageway	8 lanes (divided)	0.20	20% on the most heavily loaded lane

Design traffic

The design traffic is considered in **terms of the cumulative number of standard axles in the lane carrying maximum traffic during the design life of the road**. This can be computed using the following equation:

$$N = \frac{365 \times [(1+r)^n - 1]}{r} \times A \times D \times F$$

Where,

N is the cumulative number of standard axles to be catered for the design in terms of million standards axle (msa),

A is the initial traffic in the year of completion of construction in terms of the number of commercial vehicles per day,

D is the lane distribution factors,

F is the vehicle damage factor,

n is the design life in years, and

R is the annual growth rate of commercial vehicles (=-0.075 if growth rate is 7.5 percent per annum).

The traffic in the year of completion is estimated using the following formula:

$$A = P(1+r)^x$$

where P is the number of commercial vehicles as per last count, and x is the number of ears between the last count and the year of completion between the last count and the year of completion of the project.

Numerical example1-

Qu. Design the pavement for construction of a new bypass with the following data:

1.Two lane carriage way

2. Initial traffic in the year of completion of construction = 400 CVPD (sum of both directions)

3.Traffic growth rate = 7.5 %

4.Design life = 15 years

5. Vehicle damage factor based on axle load survey = 2.5 standard axle per commercial vehicle

6.Design CBR of subgrade soil = 4%.

Solution

1. Distribution factor = 0.75

2.

$$N = \frac{365 \times [(1+0.075)^{15} - 1)]}{0.075} \times 400 \times 0.75 \times 2.5$$

$$= 7200000$$

$$= 7.2\ msa$$

3. Total pavement thickness for CBR 4% and traffic 7.2 msa from IRC:37 2001 chart1 = 660 mm

4. Pavement composition can be obtained by interpolation from Pavement Design Catalogue (IRC:37 2001).

(a) Bituminous surfacing = 25 mm SDBC + 70 mm DBM

(b) Road-base = 250 mm WBM

(c) sub-base = 315 mm granular material of CBR not less than 30 %

Numerical example 2-

Qu. Design the pavement for construction of a new two lane carriageway for design life 15 years using IRC method.

1. The initial traffic in the year of completion in each direction is 150 CVPD and growth rate is 5%.

2. Vehicle damage factor based on axle load survey = 2.5 std axle per commercial vehicle.

3. Design CBR of subgrade soil=4%.

Solution

1. Distribution factor = 0.75

2. $$N = \frac{365 \times [(1+0.05)^{15} - 1)]}{0.05} \times 300 \times 0.75 \times 2.5$$
$$= 4430348.837$$
$$= 4.4\ msa$$

3. Total pavement thickness for CBR 4% and traffic 4.4 msa from IRC:37 2001 chart1 = 580 mm

4. Pavement composition can be obtained by interpolation from Pavement Design Catalogue (IRC:37 2001).

(a) Bituminous surfacing = 20 mm PC + 50 mm BM

(b) Road-base = 250 mm Granular base

(c) sub-base = 280 mm granular material.

SUMMARY-

- The design procedure given by IRC makes use of the CBR value, million standard axle concept, and vehicle damage factor.
- Traffic distribution along the lanes are taken into account. The design is meant for design traffic which is arrived at using a growth rate.

Design of rigid pavement using IRC: 58-2011

Overview-

- As the name implies, **rigid pavements are rigid i.e, they do not flex much** under loading like flexible pavements. They are constructed using cement concrete.
- In this case, the load carrying capacity is mainly due to the **rigidity and high modulus of elasticity** of the slab (slab action).
- **H. M. Westergaard** is considered the pioneer in providing the rational treatment of the rigid pavement analysis.

S.No	Step/Method	Description
1	**Design Approach**	Mechanistic-Empirical Design (based on stresses and fatigue analysis)
2	**Traffic Analysis**	Design for cumulative number of commercial vehicles (in million standard axles, msa) over design life
3	**Design Axle Load**	Consider single, tandem, and tridem axle load configurations
4	**Design Parameters**	- Modulus of Subgrade Reaction (k) - Elastic modulus of concrete - Poisson's ratio - Flexural strength of concrete (modulus of rupture)
5	**Critical Stress Conditions**	- Load-induced stress at slab bottom - Temperature warping stress
6	**Fatigue Analysis**	Calculate stress ratio = (flexural stress / flexural strength) and check cumulative fatigue damage (CFD ≤ 1)
7	**Slab Thickness Design**	Choose slab thickness so that total stress (load + temperature) is less than permissible flexural strength
8	**Load Transfer at Joints**	Design for dowel bars or aggregate interlock (depending on traffic)
9	**Spacing of Joints**	Determine longitudinal and transverse joint spacing to control cracking
10	**Sub-base Design**	Provide dry lean concrete (DLC) or granular sub-base layer below concrete slab
11	**Shoulder and Edge Support**	Provide tied concrete shoulders for edge load support
12	**Temperature Effects**	Consider daily and seasonal temperature variations for warping stresses

Steps in Design of Rigid Pavement (IRC:58-2011)

1. Selection of Design Parameters
• Traffic data (commercial vehicles per day, CVPD)
• Design life (typically 30 years)
• Cumulative number of standard axle load repetitions (msa)

2. Estimation of Design Traffic

3. Subgrade Characterization
• Determine modulus of subgrade reaction (k) via plate load test or CBR correlation
• Use modified k-value considering sub-base or drainage layer

4. Determination of Fatigue Life
5. Computation of Stresses

• Use Westergaard's equations to compute:
  - Edge stress - Corner stress - Interior stress

6. Thickness Design
• Select trial thickness
• Recalculate stresses and verify fatigue life
• Revise thickness if needed

7. Check for Corner Failure
• Compute corner stress and check against permissible limits

• Finalize slab thickness, joints, reinforcement, and support layers

8. Design for Temperature Stresses
• Compute warping stress and consider combined stress

9. Joint Design
• Expansion, contraction, and longitudinal joints
• Decide spacing and reinforcement (if needed)

10. Design of Dowel Bars and Tie Bars
• Design based on wheel load transfer
• Check bearing stress, bending, and shear

11. Shoulder and Sub-base Design
• Provide suitable shoulders
• Design of DLC sub-base and drainage layer

12. Finalization of Pavement Components

Modulus of sub-grade reaction-

- Westergaard considered the rigid pavement slab as a thin elastic plate resting on soil sub- grade, which is assumed as a dense liquid.
- The upward reaction is assumed to be proportional to the deflection.
- Based on this assumption, Westergaard defined a modulus of sub-grade reaction K in kg/cm3 given by

$$K = p / \text{deflection}$$

Where, is the displacement level taken as 0.125 cm and p is the pressure sustained by the rigid plate of 75 cm diameter at a deflection of 0.125 cm.

Radius of relative stiffness, *l*

(Westergaard's slab–foundation theory for concrete pavements)

Symbol	Meaning	Typical range (SI units)
E	Elastic modulus of concrete	25–40 GPa
h	Slab thickness	0.15–0.35 m
v	Poisson's ratio of concrete	0.15–0.20
k	Modulus of sub-grade reaction	20–150 MN/m³

$$l = \left[\frac{E\,h^3}{12\,k\,(1-\nu^2)}\right]^{\frac{1}{4}}$$

* *l* is the characteristic length at which a loaded concrete slab "feels" the support from the underlying soil or base.

What it represents

•It controls stress and deflection distributions under wheel loads, temperature curling, etc.

•Bigger *l* ⇒ slab behaves more like a thick, flexible plate (deflection spread over a wider area).

•Smaller *l* ⇒ support is stiffer and deflection is more localized.

Practical use

1.Corner, edge, and interior stress formulas – all contain terms like *x/l* or *y/l*.

2.Load-transfer devices & joint spacing – designs target certain *l* values to balance slab thickness vs. dowel size/spacing.

3.Warping/curling analysis – thermal and moisture gradients use *l* to compute restraint stresses.

Critical load positions-

- Since the pavement slab has finite length and width, either the character or the intensity of maximum stress induced by the application of a given traffic load is dependent on the location of the load on the pavement surface.
- There are three typical locations namely the interior, edge and corner, where differing conditions of slab continuity exist.
- **These locations are termed as critical load positions.**

WHEEL LOAD STRESSES - WESTERGAARD'S STRESS EQUATION

- The cement concrete slab is assumed to be homogeneous and to have uniform elastic properties with vertical sub-grade reaction being proportional to the deflection.
- Westergaard developed relationships for the stress at interior, edge and corner regions, denoted as σi, σe, σc in kg/cm2 respectively and given by the equation.

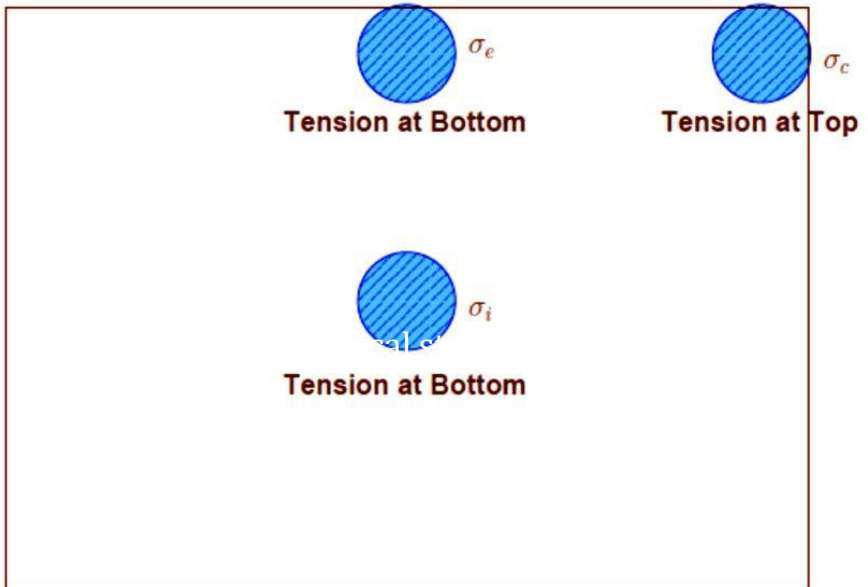
$$\sigma_i = \frac{0.316\ P}{h^2}\left[4\ \log_{10}\left(\frac{l}{b}\right) + 1.069\right]$$

$$\sigma_e = \frac{0.572\ P}{h^2}\left[4\ \log_{10}\left(\frac{l}{b}\right) + 0.359\right]$$

$$\sigma_c = \frac{3\ P}{h^2}\left[1 - \left(\frac{a\sqrt{2}}{l}\right)^{0.6}\right]$$

Where,

- h is the slab thickness in cm,
- P is the wheel load in kg,
- a is the radius of the wheel load distribution in cm,
- l the radius of the relative stiffness in cm and
- b is the radius of the resisting section in cm.



Temperature stresses-

Temperature stresses are developed in cement concrete pavement **due to variation in slab temperature**.
(i) daily variation resulting in a temperature gradient across the thickness of the slab and
(ii) seasonal variation resulting in overall change in the slab temperature.

Warping stress-

The warping stress at the interior, edge and corner regions, denoted as σi, σe, σc in kg/cm2 respectively and given by the equation-

$$\sigma_{t_i} = \frac{E\epsilon t}{2}\left(\frac{C_x + \mu C_y}{1-\mu^2}\right) \qquad \sigma_{t_e} = \text{Max}\left(\frac{C_x E\epsilon t}{2}, \frac{C_y E\epsilon t}{2}\right)$$

$$\sigma_{t_c} = \frac{E\epsilon t}{3(1-\mu)}\sqrt{\frac{a}{l}}$$

E is the modulus of elasticity of concrete in kg/cm^2 (3x10^5), e is the thermal coefficient of concrete per Degree Celsius (1x10^-7) t is the temperature difference between the top and bottom of the slab,
Cx and Cy are the coefficient based on Lx/l in the desired direction and Ly/l right angle to the desired direction, is the Poisson's ration (0.15), a is the radius of the contact area and l is the radius of the relative stiffness.

Frictional stresses-

The frictional stress σf in kg/cm^2 is given by the equation.

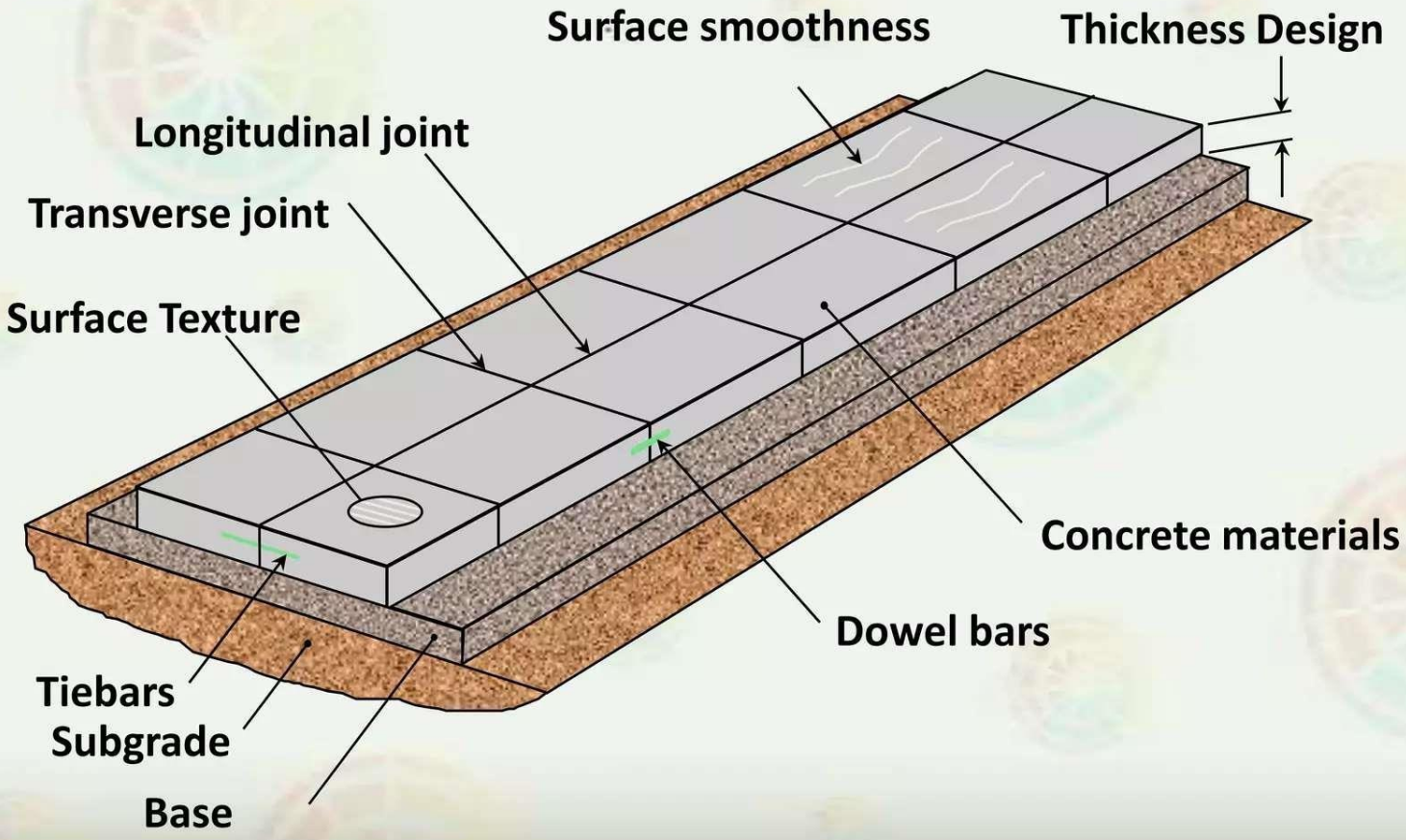
$$\sigma_f = \frac{WLf}{2 \times 10^4}$$

W is the unit weight of concrete in kg/cm^2 (2400),
f is the coefficient of sub grade friction (1.5) and
L is the length of the slab in meters.

Combination of stresses-

- The cumulative effect of the different stress give rise to the following thee critical cases
- **Summer, mid-day**: The critical stress is for edge region given by $\sigma critical = \sigma e + \sigma te - \sigma ef$
- **Winter, mid-day**: The critical combination of stress is for the edge region given by $\sigma critical = \sigma e + \sigma te + \sigma f$
- **Mid-night**s: The critical combination of stress is for the corner region given by $\sigma critical = \sigma c + \sigma tc$

Basic Components of a Concrete Pavement



Design of joints-

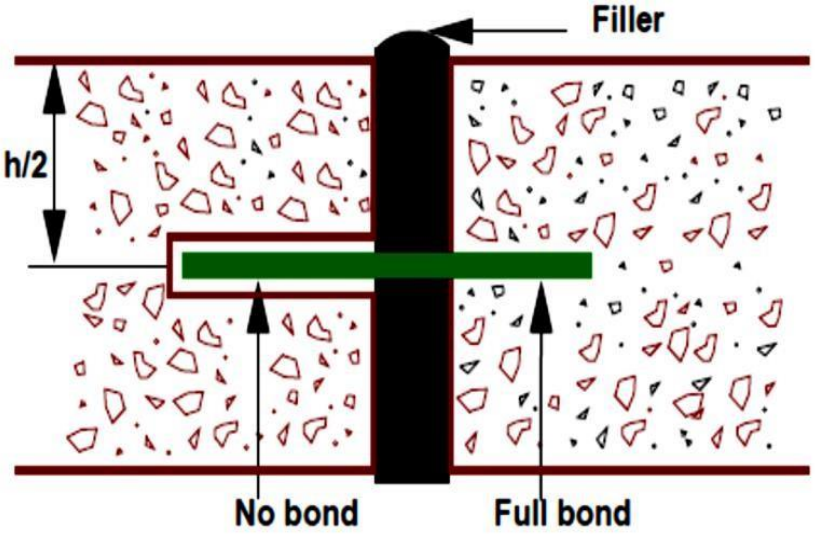
Expansion joints-

The purpose of the expansion joint is to allow the expansion of the pavement due to a rise in temperature with respect to construction temperature.

The design considerations are:

Provided along the longitudinal direction,

Design involves finding the joint spacing for a given expansion joint thickness (say 2.5 cm specified by IRC) = subjected to some maximum spacing (say 140 as per IRC)



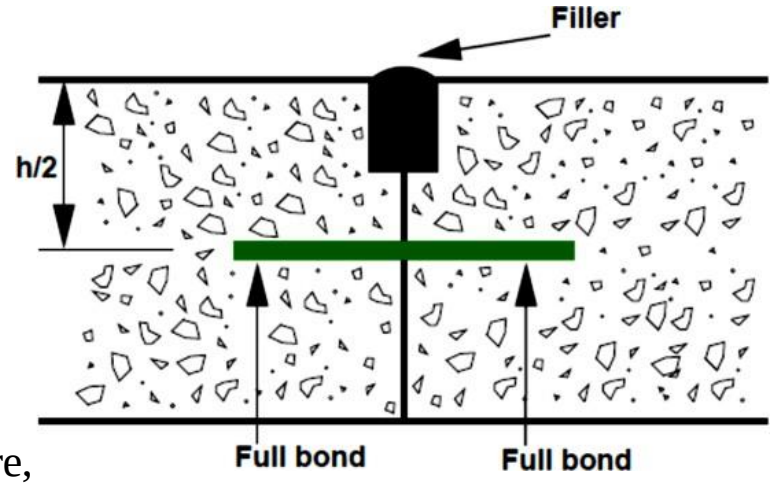
Contraction joints-

The purpose of the contraction joint is to allow the contraction of the slab due to fall in slab temperature below the construction temperature.

The design considerations are:

- The movement is restricted by the sub-grade friction
- Design involves the length of the slab given by:

$$L_c = \frac{2 \times 10^4 S_c}{W.f}$$



Where,

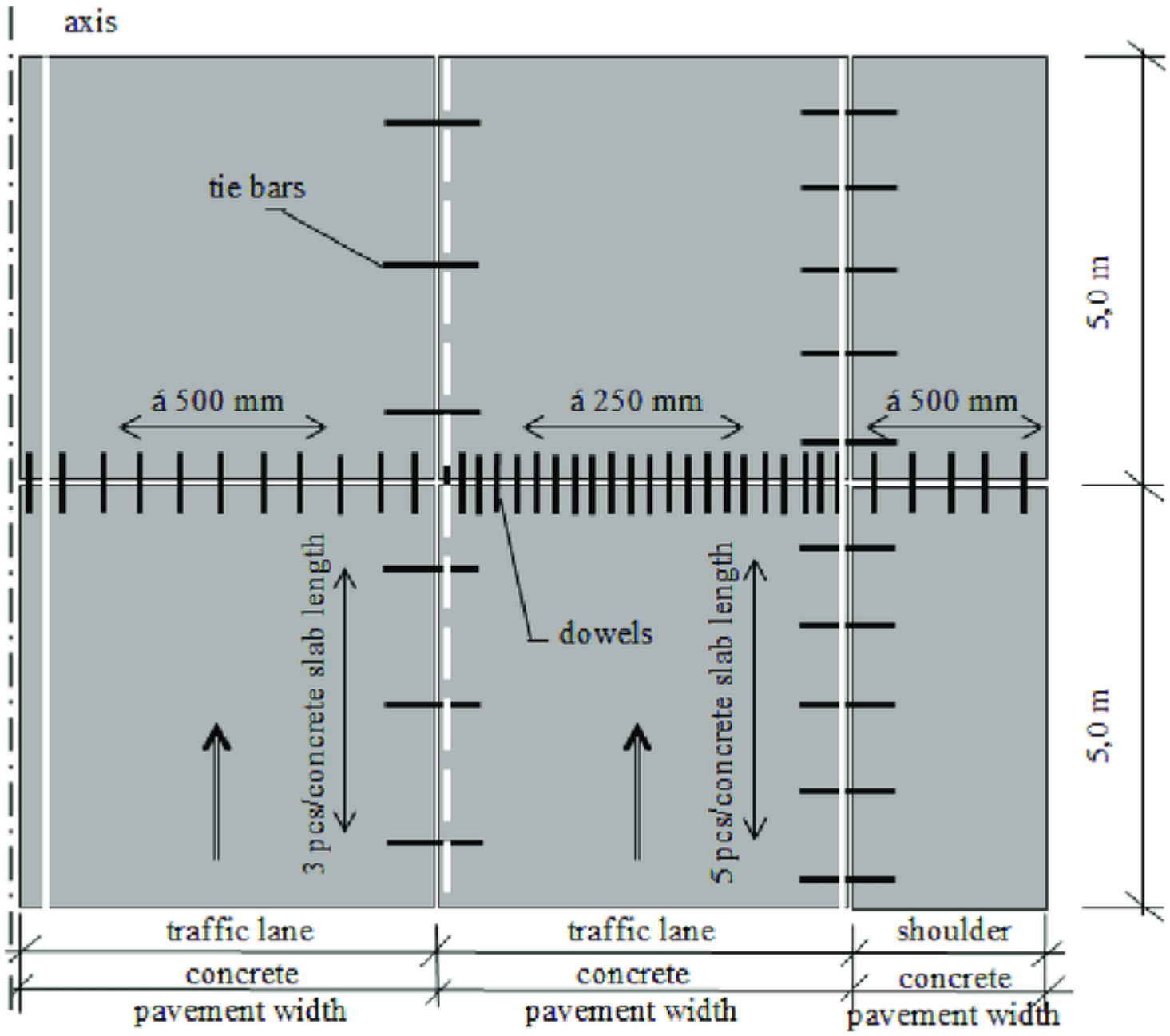
- Sc is the allowable stress in tension in cement concrete and is taken as 0.8 kg/cm^2,
- W is the unit weight of the concrete which can be taken as 2400 kg/cm^3 and
- f is the coefficient of sub-grade friction which can be taken as 1.5.

Dowel bars

- The purpose of the dowel bar is to **effectively transfer the load between two concrete slabs** and to keep the two slabs in same height.
- The dowel bars are **provided in the direction of the traffic (longitudinal**).
- The design considerations are:
  - **Mild steel rounded bars**,
  - Bonded on one side and free on other side

Tie bars

- In contrast to dowel bars, tie bars are not load transfer devices, but **serve as a means to tie two slabs**.
- Hence tie bars must be deformed or hooked and must be **firmly anchored into the concrete to function properly**.
- They are **smaller than dowel bars** and placed at **large intervals**.
- They are provided across longitudinal joints.


axis
tie bars
á 500 mm
á 250 mm
á 500 mm
5,0 m
5,0 m
dowels
3 pcs/concrete slab length
5 pcs/concrete slab length
traffic lane
concrete
pavement width
traffic lane
concrete
pavement width
shoulder
concrete
pavement width

Dowel bars Design procedure

Step 1 Find the length of the dowel bar embedded in slab Ld by equating.

Step 2 Find the load transfer capacities Ps, Pf , and Pb of single dowel bar with the Ld.

$$L_d = 5d\sqrt{\frac{F_f}{F_b}\frac{(L_d + 1.5\delta)}{(L_d + 8.8\delta)}}$$

Step 3 Assume load capacity of dowel bar is 40 percent wheel load, find the load capacity factor f as

$$\max\left\{\frac{0.4P}{P_s}, \frac{0.4P}{P_f}, \frac{0.4P}{P_b}\right\}$$

Step 4 Spacing of the dowel bars.

- Effective distance up-to which effective load transfer take place is given by 1:8 *l*,

where l is the radius of relative stiffness.

- Assume a linear variation of capacity factor of 1.0 under load to 0 at 1:8 *l*.
- Assume a dowel spacing and find the capacity factor of the above spacing.
- Actual capacity factor should be greater than the required capacity factor.
- If not, do one more iteration with new spacing.

Tie bars Design procedure-

Step 1 Diameter and spacing: The diameter and the spacing is first found out by equating the total sub-grade friction tot he total tensile stress for a unit length (one meter). Hence the area of steel per one meter in cm^2 is given by:

where,

b is the width of the pavement panel in m, h is the depth of the pavement in cm,

W is the unit weight of the concrete (assume 2400 kg=cm2),

f is the coefficient of friction (assume 1:5), and

Ss is the allowable working tensile stress in steel (assume 1750 kg=cm2)

$$A_s \times S_s = b \times h \times W \times f$$

$$A_s = \frac{bhWf}{100 S_s}$$

Assume 0:8 to 1:5 cm bars for the design.

Step 2 Length of the tie bar: Length of the tie bar is twice the length needed to develop bond stress equal to the working tensile stress and is given by:

where,

d is the diameter of the bar,

Ss is the allowable tensile stress in kg=cm2, and

$$L_t = \frac{d\ S_s}{2\ S_b}$$

Sb is the allowable bond stress and can be assumed for plain and deformed bars respectively as 17:5 and 24:6 kg=cm2.

Other modes of Transport

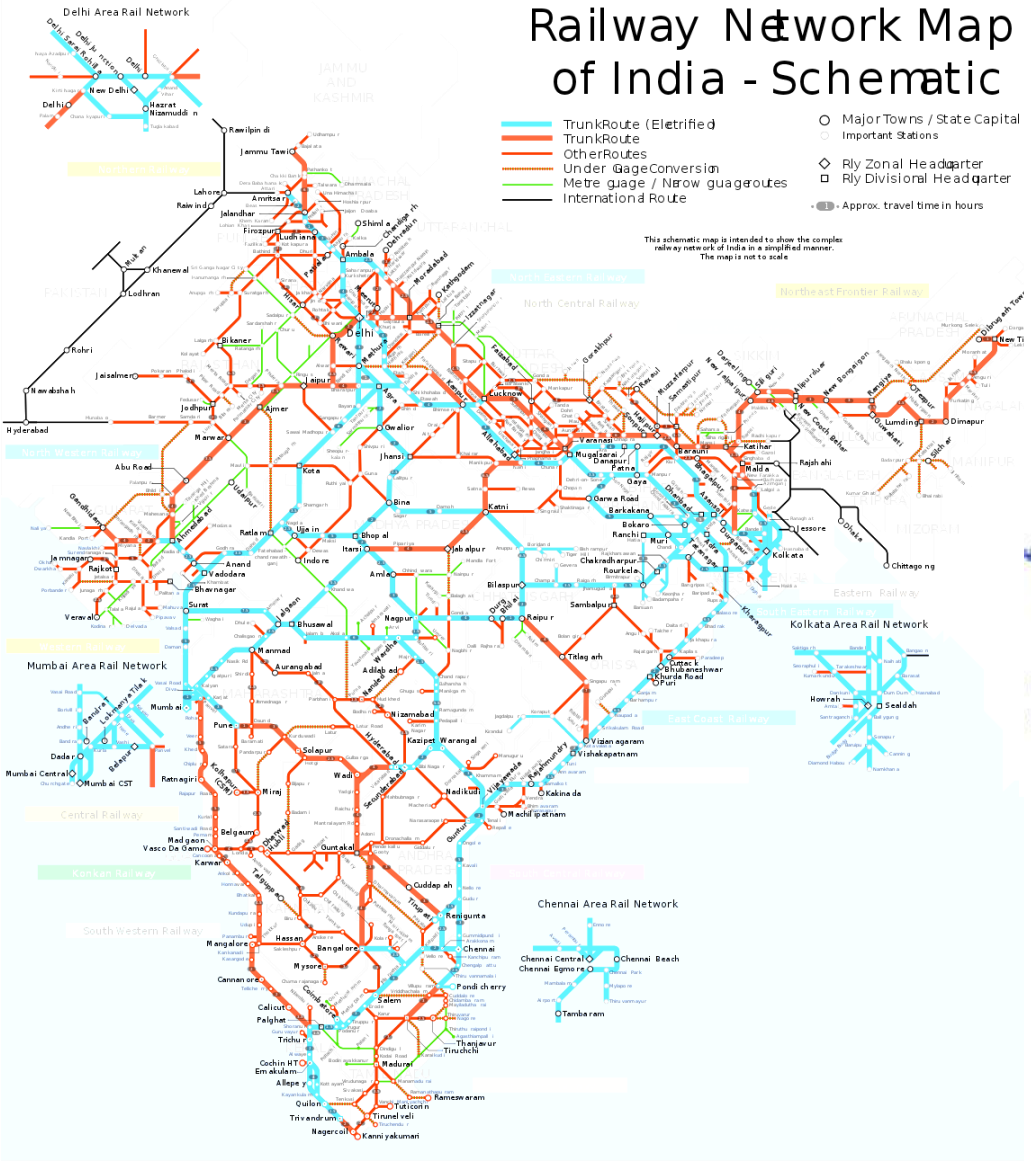


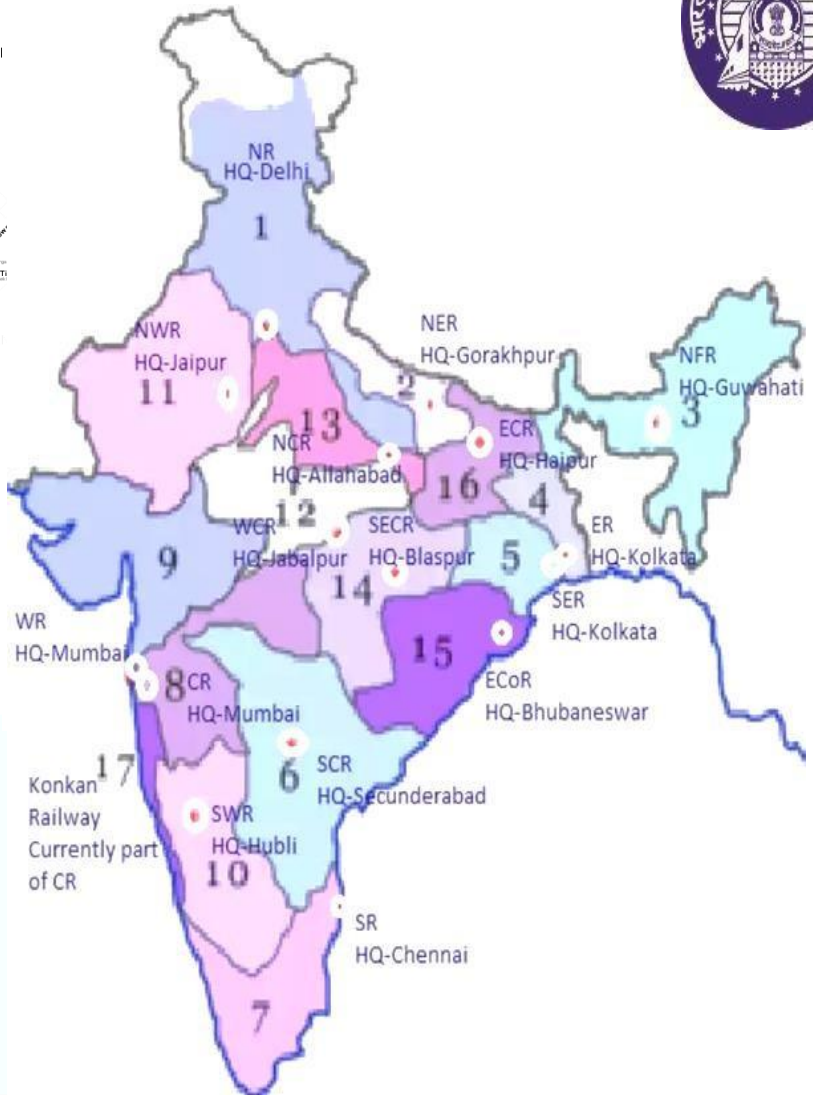
Introduction to Railways

- Indian Railways (IR) is India's national railway system operated by the Ministry of Railways.
- It is run by the government as a public, goods and manages the fourth largest railway network in the world by size, with a route length of **95,981 km** (59,640 mi) as of March 2019.
- About 63% of all the routes are electrified with 25 kV 50 Hz AC electric traction.
- In the fiscal year ending March 2019, IR carried **8.44 billion passengers** and transported **1.23 billion tonnes of freight.**
- IR runs **13,523 passenger trains daily**, on both long-distance and suburban routes, covering **7,321 stations** across India.
- In the freight segment, IR runs **more than 9,146** trains daily.

- ✓ Indian Railways
- ✓ Type- Government
- ✓ Industry- Rail transport
- ✓ Founded- 8 May 1845 (175 years ago)
- ✓ Headquarters- New Delhi, India
- ✓ Area served
- ✓ Key people
  - ❑ Piyush Goyal (Minister of Railways)
  - ❑ Suresh Angadi (Minister of State for Railways)
  - ❑ Vinod Kumar Yadav (Chairman, Railway Board)
- ✓ Services-
  - ➢ Passenger railways
  - ➢ Freight services
  - ➢ Parcel carrier
  - ➢ Catering and tourism services
  - ➢ Parking lot operations
  - ➢ Other related services
- ✓ Revenue Increase ₹1.97214 trillion (2018–19)
- ✓ Owner Government of India (100%)
- ✓ Number of employees- 1.23 million (2019)
- ✓ Divisions- 18 zones
- ✓ Subsidiaries- CRIS, CONCOR, RITES, IRCON, IRCTC,
  DFCCI, RailTel, MRVC, RVNL, NHSRCL, IRFC

Introduction to Railways





1. The first train in India ran on 16th April 1853 from Bombay to Thane.
2. The full track length of Indian Railways can circle the equator one-and-a-half times.
3. The first railroad was constructed by two Indians – Jaganath Shunkerseth and Jamsetjee Jeejeebhoy.
4. Metupalayam Ooty Nilgiri Passenger train is the slowest train in India. It runs at a speed of 10 kmph.
5. The total distance covered by Indian Railways daily equals three and a half times the distance to the moon.
6. With over 1.6 million employees, Indian Railways is the world's 9th largest employer.
7. Howrah-Amritsar Express has 115 stops, the record maximum for an Express train.
8. Indian Railways carry more than 25 million passengers every day, more than the entire population of Australia.
9. With a length of 1366.33 metres, Gorakhpur in Uttar Pradesh is the longest platform in the world.
10. Indian Railways is one of the world's largest railway networks, with 115,000 kms of tracks and a route of 65,000 kms.
11. The world's highest railway bridge is being built over Chenab. It will dwarf the Eiffel Tower.

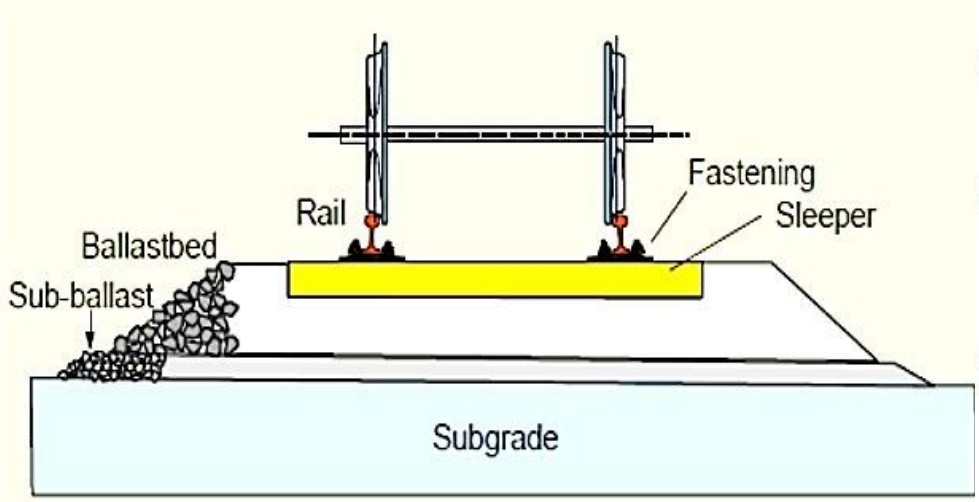
Railway Engineering:

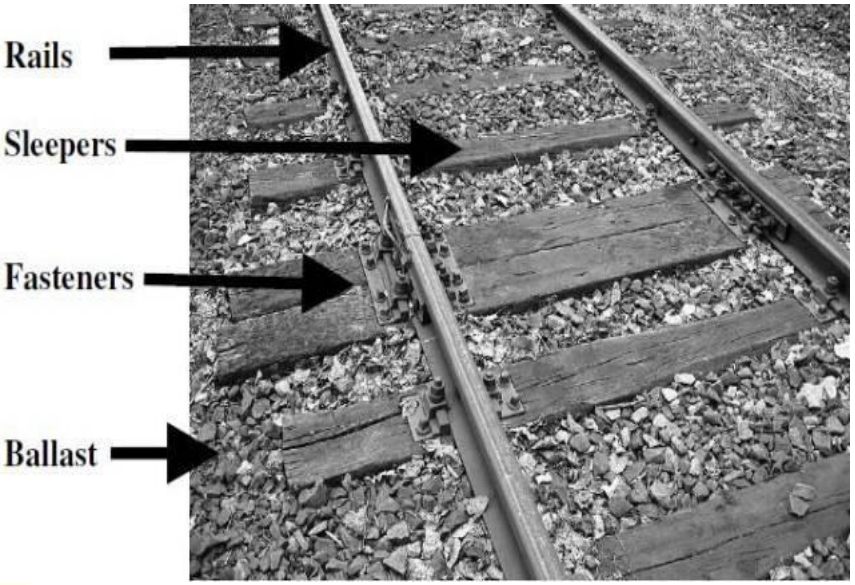
Railway engineering is a multi-faceted engineering discipline dealing with the design, construction and operation of all types of rail transport systems.

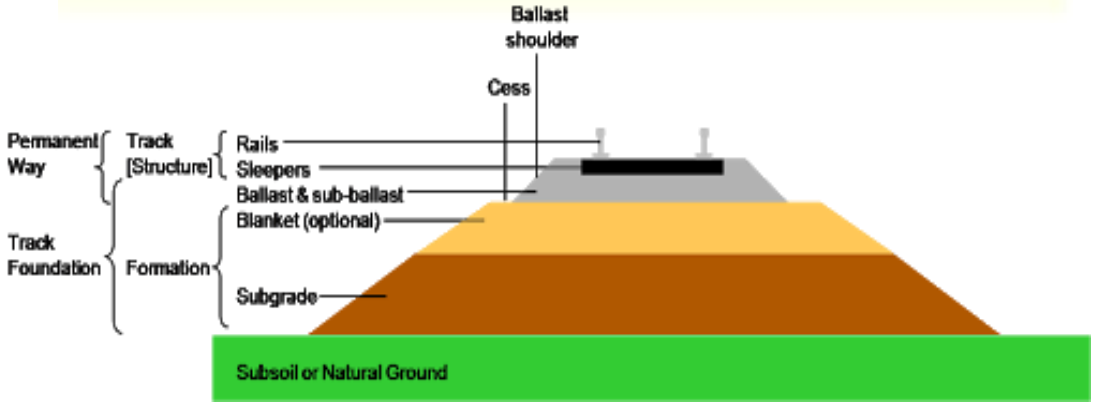
It encompasses a wide range of engineering disciplines, including-

- Civil Engineering,
- Computer Engineering,
- Electrical Engineering,
- Mechanical Engineering,
- Industrial Engineering and
- Production Engineering.

Definition :- The combination of rails, fitted on sleepers and resting on ballast & sub grade is called the railway track



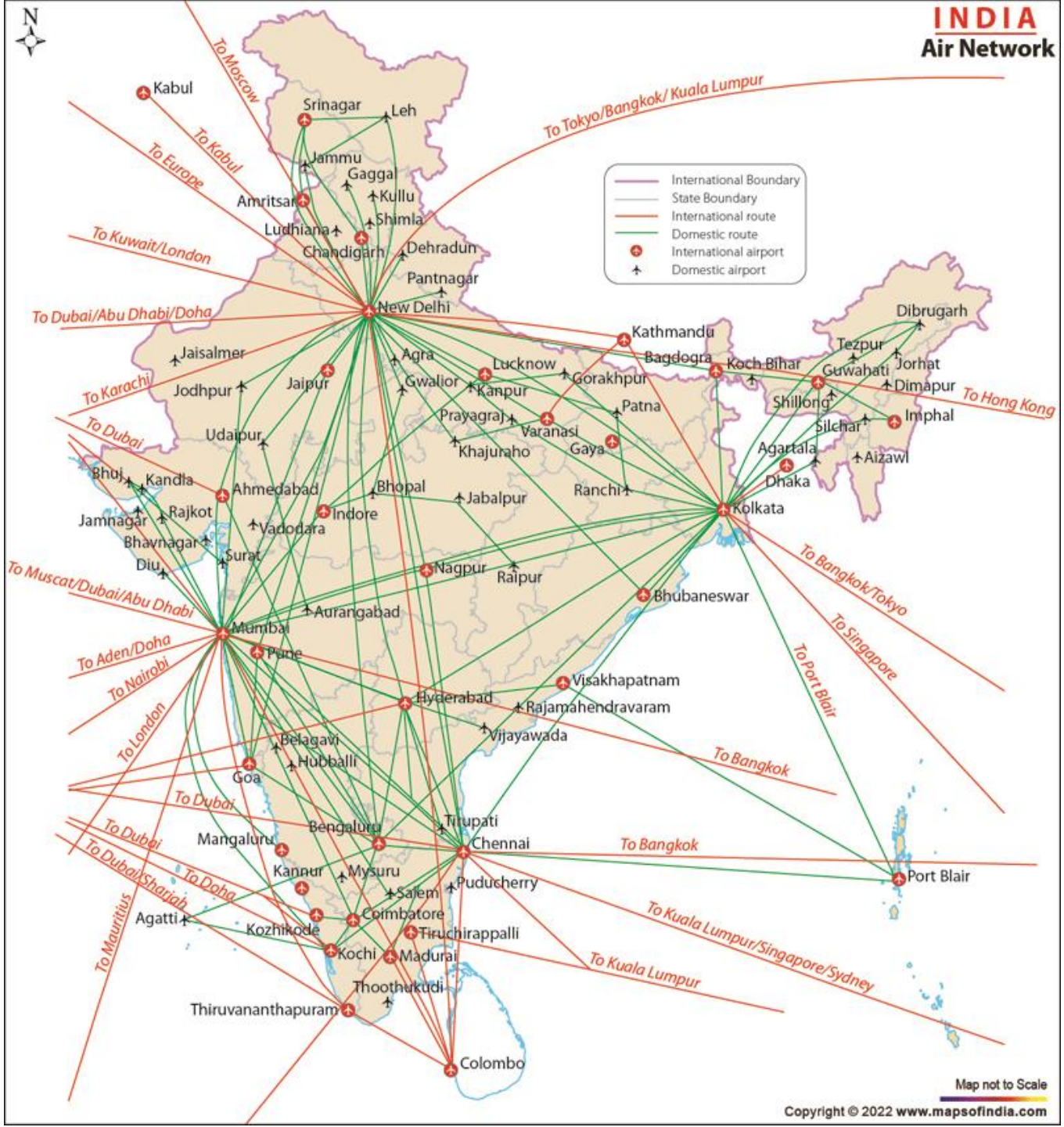




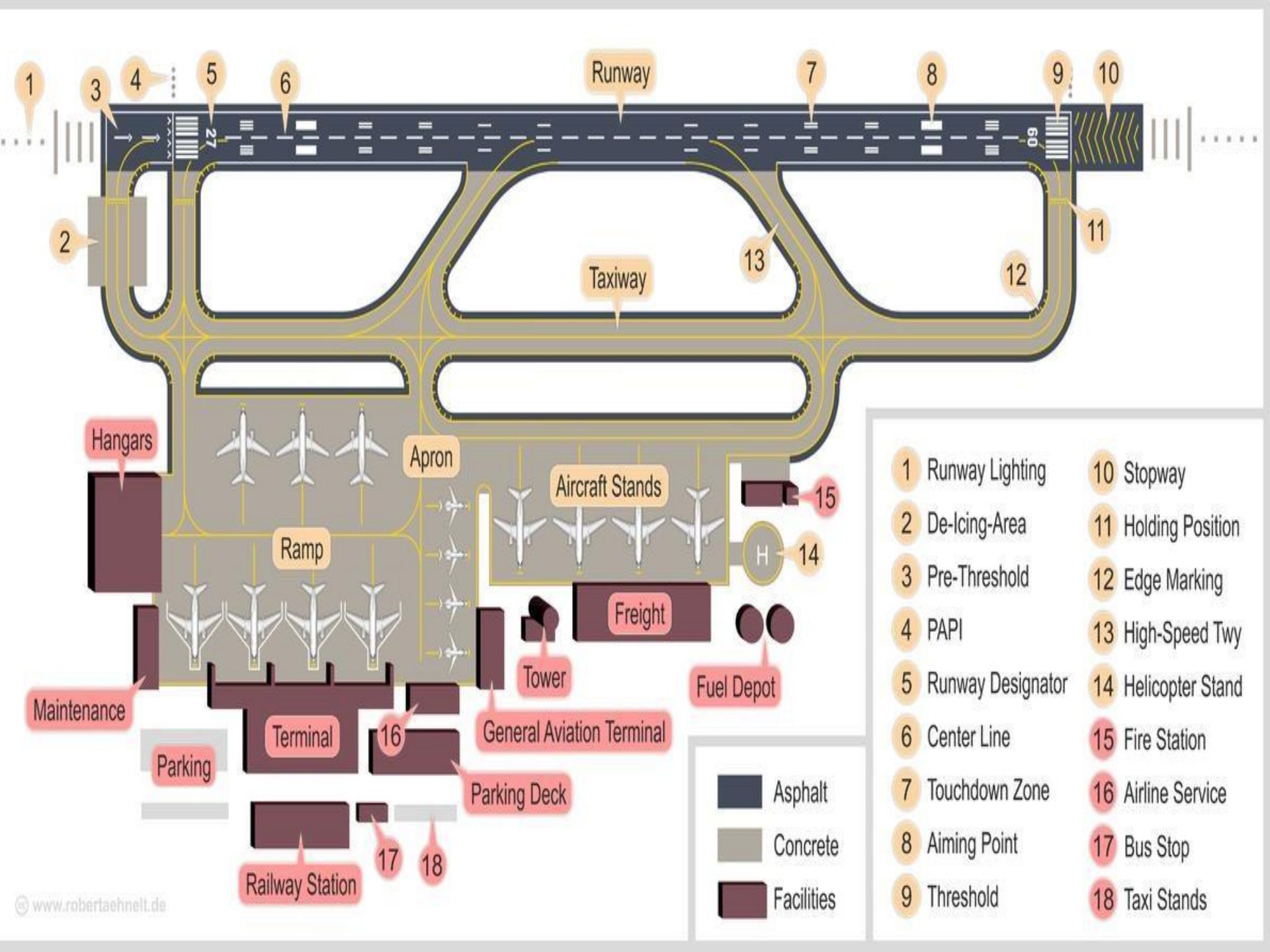
Airways

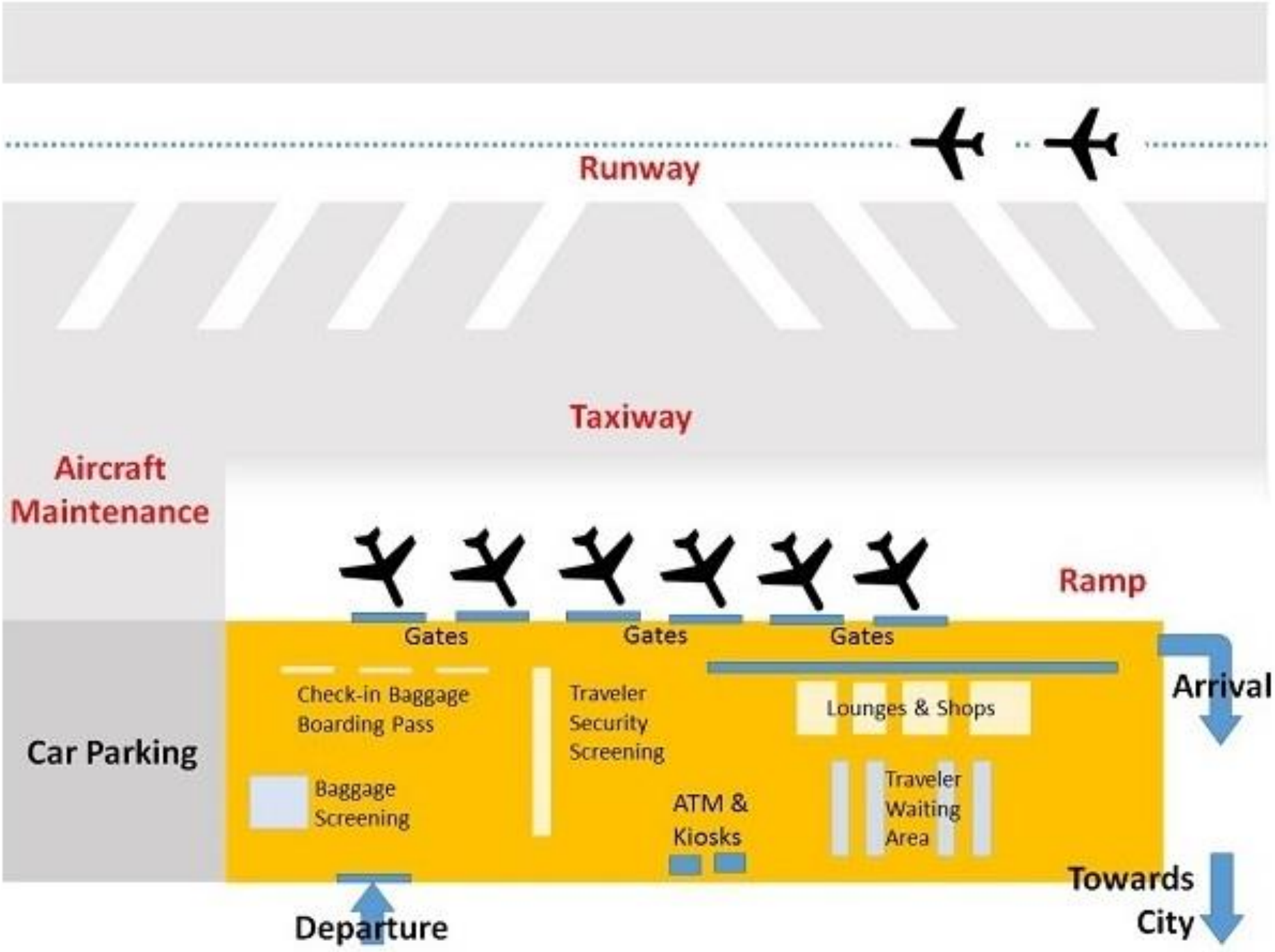
Topic	Details
Definition	Airways are routes in the sky that aircraft use to travel, similar to roads on the ground.
Importance	- Fast travel over long distances - Connects cities and countries globally - Supports emergencies, business, and tourism
Key Parts	- Airports (for takeoff, landing, services) - Air Traffic Control (guides flights) - Flight Routes (predefined sky paths) - Aircraft
Advantages	- Saves time - High safety standards - Access to remote areas
Challenges	- High cost - Weather-related delays - Requires advanced maintenance and security

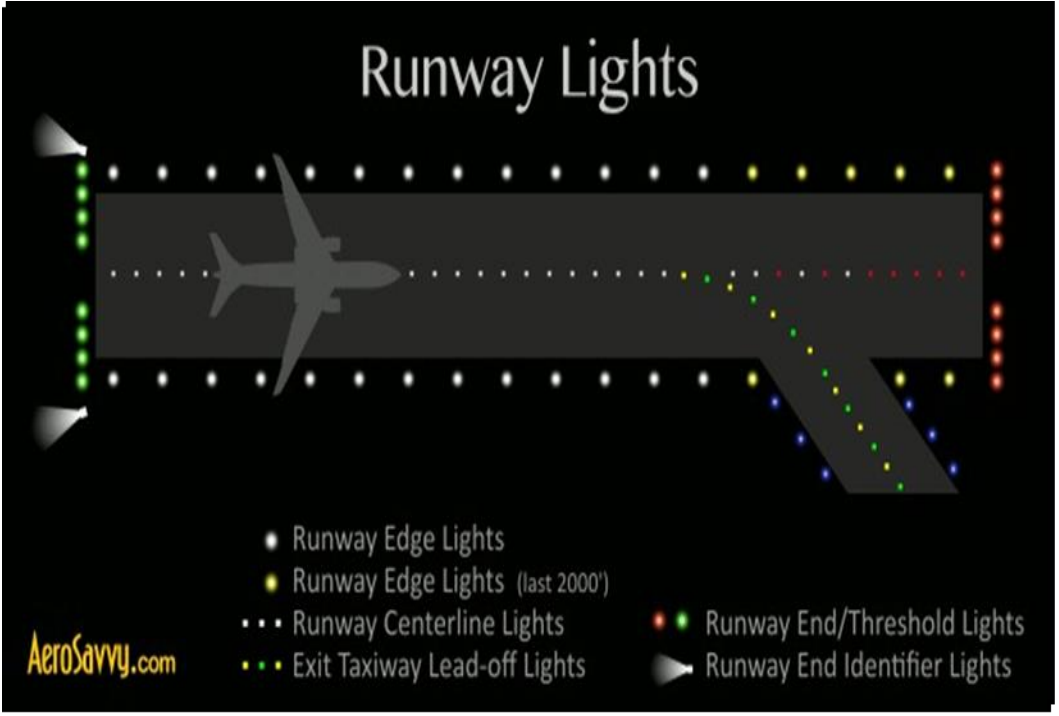
Beginning of Air Transport	Started in 1911 with airmail between Allahabad and Naini (10 km).
Development	Major growth after Independence; Airport Authority of India (AAI) manages air traffic and 125 airports.
Nationalisation	In 1953, air transport was nationalized into Air India International (now Air India) and Indian Airlines (now 'Indian').
Private Airlines	Many private and low-cost carriers now dominate the industry, making air travel affordable.
Key Organizations	- **Airport Authority of India (AAI)**: Manages airports, airspace. - **Pawan Hans Limited**: Helicopter services for hilly and petroleum sectors.
History Timeline	- 1911: First flight (Allahabad-Naini). - 1947: Major companies - Indian National Airways, Tata Sons, Air Services of India, Deccan Airways. - 1951: More companies - Bharat Airways, Himalayan Aviation, Airways India, Kalinga Airlines. - 1953: Nationalization.
Air India Services	Provides international passenger and cargo services; connects all continents.
2010 Statistics	- 520.21 lakh passengers - 23 lakh metric tonnes of cargo.
Directorate General of Civil Aviation (DGCA)	Responsible for: - Registration, licensing, certification - Regulation of air traffic - Accident investigations - Environmental regulations - Promotion of indigenous aircraft manufacturing
Other Functions (DGCA)	- Amendments to Aircraft Act - Coordination with ICAO - Flexi-use of airspace - Safety programs - Carriage of dangerous goods.



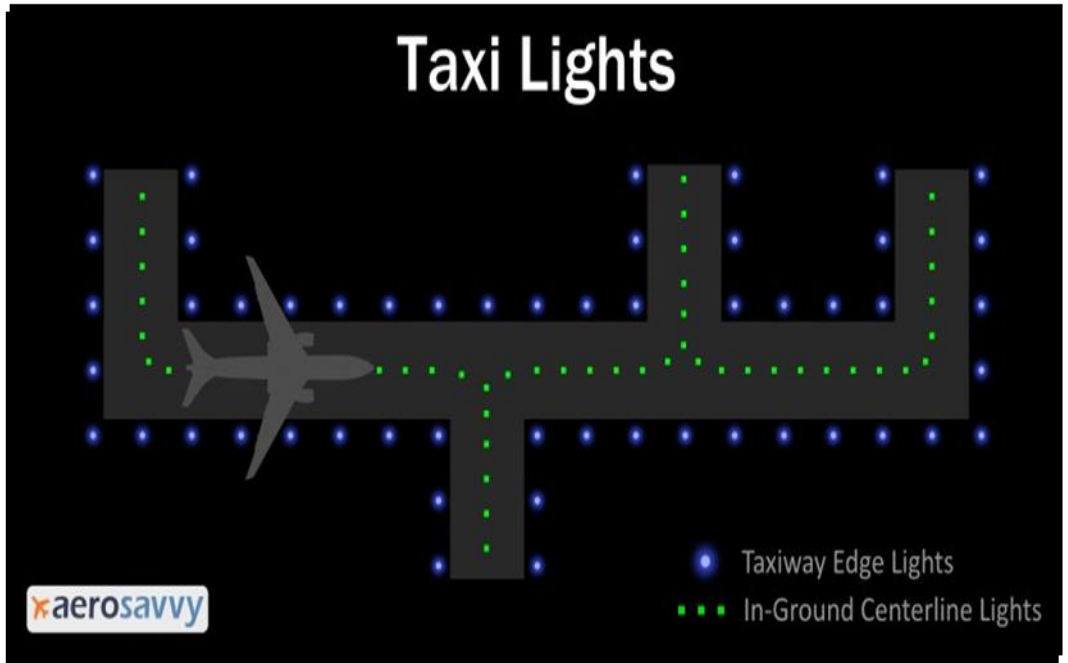
Copyright © 2022 www.mapsofindia.com


1
3
4
5
6
Runway
7
8
9
10
27
09
2
11
13
12
Taxiway
Hangars
Apron
Aircraft Stands
15
Ramp
H
14
Freight
Tower
Fuel Depot
Maintenance
Terminal
16
General Aviation Terminal
Parking
Parking Deck
17
18
Railway Station
1 Runway Lighting
2 De-Icing-Area
3 Pre-Threshold
4 PAPI
5 Runway Designator
6 Center Line
7 Touchdown Zone
8 Aiming Point
9 Threshold
10 Stopway
11 Holding Position
12 Edge Marking
13 High-Speed Twy
14 Helicopter Stand
15 Fire Station
16 Airline Service
17 Bus Stop
18 Taxi Stands
Asphalt
Concrete
Facilities
© www.robertaehnelt.de


Runway
Taxiway
Aircraft Maintenance
Ramp
Gates
Gates
Gates
Check-in Baggage
Boarding Pass
Traveler
Security
Screening
Lounges & Shops
Arrival
Car Parking
Baggage
Screening
ATM &
Kiosks
Traveler
Waiting
Area
Towards
City
Departure


Runway Lights
Runway Edge Lights
Runway Edge Lights (last 2000')
Runway Centerline Lights
Exit Taxiway Lead-off Lights
Runway End/Threshold Lights
Runway End Identifier Lights
AeroSavvy.com




Taxi Lights
Taxiway Edge Lights
In-Ground Centerline Lights
aerosavvy

Waterways



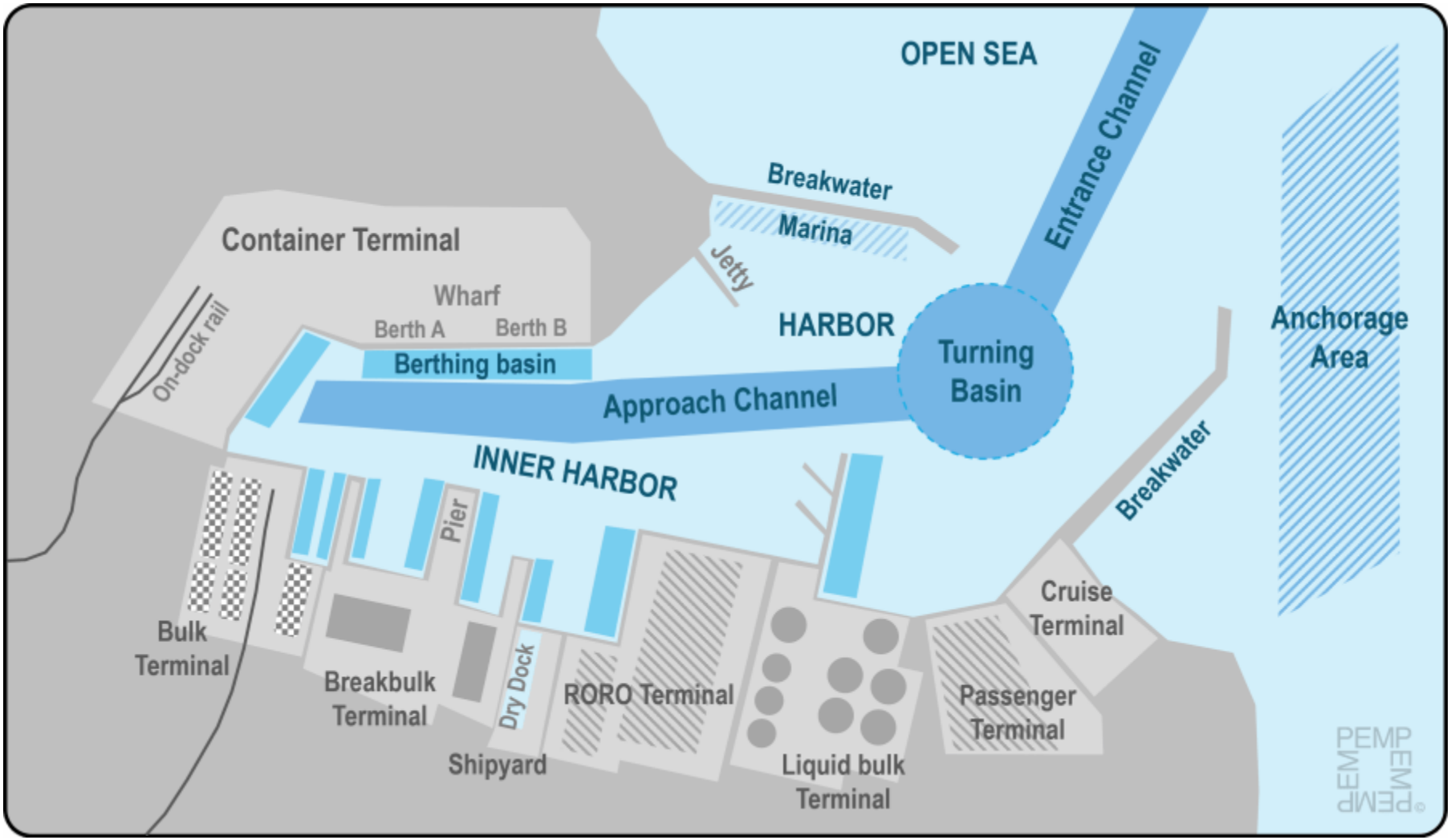
Waterways



Dock and Harbor Engineering:

Aspect	Details
Definition	Branch of civil engineering dealing with design, construction, and maintenance of docks, harbors, and related facilities for maritime activities.
Objectives	Provide safe berthing, loading/unloading, ship repair facilities; protect ships from waves; ensure efficient maritime trade and naval operations.
Types of Harbors	- **Natural Harbor** (e.g., Sydney) - **Semi-natural Harbor** (partly natural, partly man-made) - **Artificial Harbor** (e.g., Chennai Port)
Types of Docks	- **Wet Dock** (Ships afloat all the time) - **Dry Dock** (Water removed for repairs) - **Floating Dock** (Portable repair docks)
Components of a Harbor	- Entrance channel - Breakwaters - Jetties - Wharves - Quays - Piers - Mooring Dolphins - Navigation Aids
Components of a Dock	- Dock Gates - Caissons - Pumping System - Graving Dock Basin - Lock Gates (for tidal control)

Components of a Harbor



Port VS Harbour VS Dock







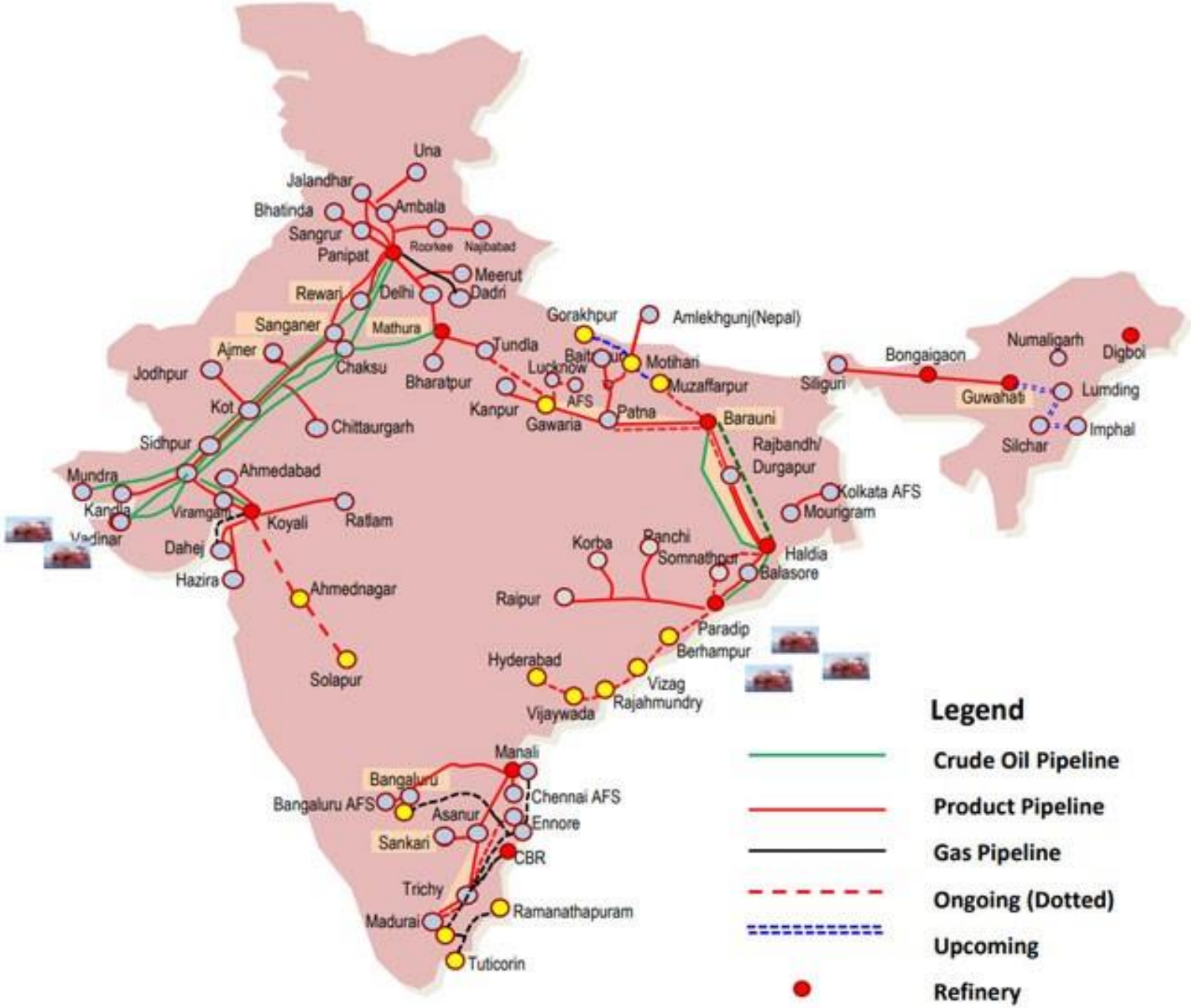
Aspect	Port	Dock	Harbour
Definition	A place where ships load and unload cargo or passengers, including complete facilities.	An enclosed area for berthing, repairing, or loading/unloading ships.	A sheltered area of water where ships can anchor safely, naturally or artificially.
Purpose	Commercial operations like cargo handling, passenger services, storage, customs, etc.	Berthing, repairing, construction, and maintenance of ships.	Provide safe anchorage, protection from rough sea and weather.
Natural/Man-made	Mostly man-made, developed with extensive infrastructure.	Entirely man-made structures.	Can be natural, semi-natural, or entirely man-made.
Typical Location	On seacoasts, river mouths, or lake shores with trade access.	Within ports or shipyards.	Along coastlines, bays, river mouths, or behind islands.
Ship Access	Ships enter for loading/unloading, registration, and customs clearance.	Ships enter for repair, construction, or specific loading/unloading.	Ships anchor temporarily or wait for port access.
Example	Port of Singapore, Port of Rotterdam, Port of Mumbai.	Dry Dock in Visakhapatnam Shipyard, Floating Dock in Cochin Shipyard.	Sydney Harbour (natural), Chennai Harbour (artificial).
Facilities Focus	Trade, customs, transportation, cargo storage.	Ship repair, ship building, maintenance facilities.	Anchoring, mooring, weather protection.
Operation Type	Commercial and administrative operations.	Mostly engineering and maintenance operations.	Nautical safety and anchorage operations.

• Inland Water Transport in India-

- India has an **extensive network** of inland waterways in the form of rivers, canals, backwaters and creeks.
- The **total navigable length is 14,500 km** (9,000 mi), out of which about 5,200 km (3,200 mi) of river and 4,000 km (2,500 mi) of canal can be used by mechanized crafts.
- About **44 million tonnes** (49,000,000 short tons) of **cargo are moved annually through these waterways** using mechanized vessels and country boats.
- Cargo transported in an organized manner is confined to a few waterways in **Goa, West Bengal, Assam and Kerala.**
- Inland waterways consist of the Ganges-Bhagirathi-Hooghly rivers, the Brahmaputra, the Barak river, the rivers in Goa, the backwaters in Kerala, inland waters in Mumbai and the deltaic regions of the Godavari- Krishna rivers.

Pipeline Transportation

- Pipelines can be used to transport all manner of substances.
- The type of material that is chosen to construct the pipe is determined by the substance it is being designed to transport.
- For example, nickel steel alloys are used very effectively in the transportation of liquified natural gas (LNG) because it is able to remain ductile even when extremely cold.
- Within the domestic sector, pipelines are used for transporting water for drinking or irrigation as well as for sewerage purposes and to convey domestic use gases to provide indoor heating or cooking facilities.
- **Pipelines ensure than there is a consistent and uninterrupted flow of vital liquids and gases to and from domestic residences**.
- The **oil industry** in particular **would not be able to operate without complex networks of pipelines as they are essential for midstream uses**, as well as for the critical transfer of finished petroleum products to end-users or dependent industries.


Una
Jalandhar
Bhatinda
Ambala
Sangrur
Roorkee
Najibabad
Panipat
Meerut
Rewari
Delhi
Dadri
Sanganer
Mathura
Gorakhpur
Amlekhgunj(Nepal)
Ajmer
Tundla
Jodhpur
Chaksu
Bharatpur
Lucknow
Motihari
Muzaffarpur
Kot
Kanpur
AFS
Patna
Barauni
Chittaurgarh
Gawaria
Sidhpur
Rajbandh/
Durgapur
Mundra
Ahmedabad
Kandla
Viramgam
Koyali
Ratlam
Kolkata AFS
Mourigram
Vadinar
Dahej
Korba
Ranchi
Somnathpur
Haldia
Balasore
Hazira
Ahmednagar
Raipur
Paradip
Berhampur
Hyderabad
Vizag
Rajahmundry
Solapur
Vijaywada
Manali
Bangaluru
Chennai AFS
Bangaluru AFS
Asanur
Ennore
Sankari
CBR
Trichy
Ramanathapuram
Madurai
Tuticorin
Numaligarh
Digboi
Bongaigaon
Siliguri
Guwahati
Lumding
Imphal
Silchar
Legend
Crude Oil Pipeline
Product Pipeline
Gas Pipeline
Ongoing (Dotted)
Upcoming
Refinery

Advantages of pipelines

There are many advantages of using pipelines to transport utility gases and liquids as opposed to other methods such as shipping, road or rail. These advantages are as follows:

- **Large Capacity:** Pipelines can **transport huge amounts of liquids and gases** and are much more efficient than conventional methods.
- **Safer:** The transportation of oil and gases obviously can be hazardous due to their occasional volatile and flammable nature. Using pipelines **minimises the risk of there being an incident during transportation**. Subterranean pipelines have very little exposure to natural elements whereas above- surface pipelines are designed to withstand the affects if adverse environments and weather.
- **Small surface footprint:** Because most pipelines are subterranean, this means that **they occupy very little above-ground surface and they are far away from populated areas.**

- **Efficient construction:** oil and gas carrier pipeline systems can be constructed and installed is a surprisingly short space of time, especially in comparison to a structure such as a rail network. This is because the pipelines can be designed to run over natural geographical barriers.
- **Lower energy use:** Pipeline systems **tend to require much lower amounts of energy to operate and therefore larger volumes of materials can be transported for much lower costs.**
- **Environmentally friendly:** pipeline transport lines are much less damaging to the environment and have a **lower carbon footprint than other methods** of transportation because they are sealed and mostly underground.

Classification, Requirements, Comparative Studies

Aspect	Railways	Airways	Waterways	Pipeline Transportation
Classification	- Passenger railways - Freight railways - Metro/Suburban railways	- Domestic flights - International flights - Cargo flights	- Inland waterways - Ocean/Sea transport - Canal transport	- Oil pipelines - Gas pipelines - Slurry pipelines
Basic Requirements	- Track network - Stations - Locomotives - Signaling system	- Airports - Runways - Air Traffic Control - Aircraft	- Ports - Docks - Ships/Boats - Navigable routes	- Network of pipes - Pumping stations - Compressor stations
Infrastructure Cost	High (laying tracks, building bridges, tunnels)	Very High (airports, air traffic systems, aircraft)	Low to moderate (natural rivers, few constructed ports)	High (pipeline installation, control systems)
Operation Cost	Moderate	Very High (fuel, maintenance, personnel)	Low (fuel-efficient, large volume transport)	Low (after installation)
Speed	Moderate (80–300 km/h)	Very High (500–900 km/h)	Low (10–50 km/h for ships)	Slow to moderate (depends on material and distance)
Capacity	Very High (huge volume of cargo/passengers)	Moderate (limited cargo/passenger space)	Very High (bulk cargo like coal, crude oil, containers)	High (continuous flow of oil, gas, etc.)
Flexibility	Low (fixed tracks)	High (flight paths can change)	Moderate (limited to waterways)	Very Low (fixed pipelines)
Suitability	Land-based long-distance transport, bulk goods, daily commuting.	Long-distance passenger travel, high-value cargo.	Heavy and bulk commodities like crude oil, coal, grains.	Continuous transport of petroleum, gas, chemicals.
Environmental Impact	Moderate (noise, land use)	High (air pollution, carbon emissions)	Low (if managed well; risk of water pollution otherwise)	Low (risk of leakage, but no emissions)